

TRANSLATIONAL SCIENCES

miR487a-3p and miR6855-3p Facilitate Macrophage Proinflammatory Polarization and Lipid Accumulation in Atherosclerosis

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BACKGROUND: The integrated regulation of microRNAs on macrophage plasticity plays a key role in atherosclerosis. We tested the hypothesis that miR487a-3p and miR6855-3p accelerate atherosclerosis by intensifying macrophage inflammatory response and metabolic dysregulation.

METHODS: The microRNA sequencing and messenger RNA sequencing were conducted in peripheral monocytes from patients with coronary artery disease (CAD) and healthy controls. Macrophages in mouse aortas and human coronary arteries were characterized using flow cytometry and immunostaining. Atherosclerosis development was evaluated in male PCSK9 (proprotein convertase subtilisin/kexin type 9)-overexpression mice harboring myeloid cell-specific deficiency of CPE (carboxypeptidase E) or RRM2 (ribonucleotide reductase regulatory subunit M2) and challenged with a high-fat diet.

RESULTS: miR487a-3p and miR6855-3p were the top microRNA candidates identified by microRNA sequencing in peripheral monocytes and validated by quantitative real-time polymerase chain reaction, with significant differences between patients with CAD and controls. Both microRNAs were lipid-inducible and secreted extracellularly. KLF5 (Krüppel-like factor 5) and IRF1 (interferon regulatory factor 1) bound to the promoter regions of miR487a-3p and miR6855-3p, respectively, to enhance their transcription. Accordingly, patients with CAD exhibited significantly elevated plasma miR487a-3p and miR6855-3p levels compared with controls, which positively correlated with blood lipid levels and Gensini score (reflecting CAD severity and prognosis). The area under the receiver operating characteristic curve (≈ 0.83 for each) supported their diagnostic accuracy. Of note, miR487a-3p and miR6855-3p were predominantly expressed in coronary arterial macrophages. The dramatic expansion of miR487a-3p⁺ and miR6855-3p⁺ macrophages and the elevated expression of both microRNAs in coronary arteries were positively associated with lesion area in patients with CAD. Mechanistically, transcriptomic analyses and functional assays revealed that elevated miR487a-3p or miR6855-3p promoted macrophage proinflammatory responses, lipid metabolic dysregulation, and foam cell formation. Conversely, inhibition of either microRNA alleviated ox-LDL (oxidized low-density lipoprotein)-induced macrophage inflammatory responses and lipid metabolic dysfunction. Moreover, conditioned medium from miR487a-3p- or miR6855-3p-overexpressing macrophages promoted endothelial cell apoptosis, whereas this effect was attenuated when endothelial cells were exposed to medium from ox-LDL-treated macrophages with microRNA inhibition. Furthermore, integration of downregulated genes from monocyte and macrophage messenger RNA sequencing with TargetScan-predicted targets identified CPE and RRM2 as targets of miR487a-3p and miR6855-3p, respectively. Direct binding was confirmed by dual-luciferase assays and microRNA pull-down. Overexpression of CPE or RRM2 partially reversed the detrimental effects of miR487a-3p and miR6855-3p, respectively, on macrophage phenotypic switching and metabolic dysregulation. Conversely, monocyte-/macrophage-specific depletion of CPE or RRM2 aggravated atherosclerosis progression in hypercholesterolemic mice by instigating macrophage inflammatory responses and lipid metabolic disturbance.

CONCLUSIONS: miR487a-3p and miR6855-3p fulfill the criteria of promising biomarkers for CAD diagnosis and prognosis. Mechanistically, they intensify inflammatory responses and disrupt lipid metabolism in macrophages, identifying both microRNAs as potential therapeutic targets for CAD.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: atherosclerosis ■ coronary artery disease ■ inflammation ■ macrophages ■ microRNAs

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Nonstandard Abbreviations and Acronyms

ABCA1	ATP binding cassette subfamily A member 1
ABCG1	ATP binding cassette subfamily G member 1
APOE	apolipoprotein E
CAD	coronary artery disease
CCL	C-C motif chemokine ligand
CCR	C-C chemokine receptor
CD	control diet
CPE	carboxypeptidase E
CSF	colony-stimulating factor
CXCL	C-X-C motif chemokine ligand
CXCR	C-X-C motif chemokine receptor
EGR1	early growth response 1
GO	Gene Ontology
GSEA	Gene Set Enrichment Analysis
HC	healthy control
HFD	high-fat diet
hMDM	human monocyte-derived macrophage
IFN	interferon
IL	interleukin
IRF1	interferon regulatory factor 1
JAK/STAT	Janus kinase/signal transduction and transcription activation
KEGG	Kyoto Encyclopedia of Genes and Genomes
KLF5	Krüppel-like factor 5
LDL-C	low-density lipoprotein-cholesterol
LDLR	low-density lipoprotein receptor
MAPK	mitogen-activated protein kinase
MMP	matrix metalloproteinase
MRC1	mannose receptor C type 1
mRNA	messenger RNA
MSR1	macrophage scavenger receptor 1
NC	negative control
NF-κB	nuclear factor-κB
OLR1	oxidized low-density lipoprotein receptor 1
OR	odds ratio
ox-LDL	oxidized low-density lipoprotein
PCSK9	proprotein convertase subtilisin/kexin type 9
RNA-seq	RNA sequencing
ROC	receiver operating characteristic
RRM2	ribonucleotide reductase regulatory subunit M2
SCARB1	scavenger receptor class B member 1
sdLDL-C	small dense low-density lipoprotein-cholesterol

TC	total cholesterol
TF	transcription factor
TIMP	tissue inhibitor of matrix metalloproteinase
TLR4	toll-like receptor 4
TNF	tumor necrosis factor
TNFSF	tumor necrosis factor superfamily
VEGF	vascular endothelial growth factor

What Are the Clinical Implications?

MicroRNAs are emerging as biomarkers and therapeutic targets for cardiovascular disease, yet few have been validated for coronary artery disease. This study identifies miR487a-3p and miR6855-3p as lipid-inducible microRNAs upregulated in peripheral monocytes and coronary plaque macrophages of patients with coronary artery disease, with their plasma levels correlating with Gensini score and exhibiting robust diagnostic accuracy. Both microRNAs promote macrophage proinflammatory response, lipid metabolic disorders, and endothelial injury by targeting CPE (carboxypeptidase E) and RRM2 (ribonucleotide reductase regulatory subunit M2), respectively, and remain independent predictors of coronary artery disease after adjusting for lipid profiles. These findings establish miR487a-3p and miR6855-3p as promising minimally invasive biomarkers for coronary artery disease diagnosis and risk stratification, and suggest that therapeutic strategies targeting these microRNAs or restoring CPE/RRM2 expression may offer novel approaches for treating atherosclerosis.

MicroRNAs are small noncoding RNA molecules present in all eukaryotic cells that regulate the expression of up to 60% of protein-coding genes by binding to complementary sequences on messenger RNAs (mRNAs), thereby suppressing mRNA translation and consequently reducing protein output.¹ Their ability to modulate mRNA stability and translational efficiency makes them important posttranscriptional regulators of diverse biological processes.² MicroRNAs play essential roles in cardiovascular development and function, and a deregulated cardiac-enriched microRNA profile contributes critically to the pathogenesis of coronary artery disease (CAD).³ Recent studies have highlighted the potential of microRNAs as promising biomarkers and therapeutic targets for CAD,^{4,5} which results from atherosclerotic occlusion of the coronary arteries and remains the leading cause of death globally and a major contributor to disability.⁶ Stable plasma levels of several microRNAs are strongly associated with CAD risk and are being explored as minimally invasive biomarkers

for CAD diagnosis and prognosis; examples include miR21, miR3149, miR122-5p, miR370, and miR208a as diagnostic markers, and miR132, miR210, miR140-3p, miR423-5p, and miR320a as prognostic markers.⁷ Moreover, modulation of a single microRNA by agomir (microRNA agonist) or antagomir (microRNA antagonist) can alter the CAD disease process; for instance, antagomir21, antagomir33, miR30 mimic, and antagomir143 significantly attenuate CAD-related cardiomyocyte death and improve cardiac function, underscoring their therapeutic potential in CAD.³

Atherosclerosis has a significant inflammatory component characterized by an imbalance between proinflammatory and regulatory signals.^{8,9} Monocytes and their descendant macrophages play central roles in disease initiation and progression.¹⁰ Triggered by inflammatory cytokines, monocytes transmigrate across the endothelium to the vascular intima, where they differentiate into macrophages and take up lipids, forming foam cells that constitute a major component of the atherosclerotic plaque.¹⁰ Plaque macrophages exhibit high heterogeneity and plasticity, which shape the evolving plaque microenvironment through excessive lipid accumulation, cytokine hyperactivation, hypoxia, apoptosis, and necroptosis. The metabolic and functional transitions of plaque macrophages in response to microenvironmental cues not only influence ongoing and subsequent inflammatory responses within lesions but also directly govern atherosclerotic progression or regression.^{11,12}

Several microRNAs have been implicated in macrophage regulation during atherosclerosis. miR124 overexpression may represent a promising therapeutic strategy for atherosclerosis and CAD by inhibiting p38 and subsequent macrophage apoptosis.¹³ Conversely, miR6516-5p contributes to the loss of protective pericardial macrophages in CAD by suppressing CD36 and macrophage lipid uptake.¹⁴ Furthermore, inhibition of miR33 and overexpression of miR155 can prevent macrophage foam cell transformation and markedly reduce atherosclerotic plaque progression.^{3,15} Although these studies highlight the critical roles of individual microRNAs in macrophage functional regulation during CAD, a systemic understanding of microRNA-mediated modulation of the pleiotropic capacity of monocytes and macrophages, the central cell types in atherosclerosis, remains lacking.

In this study, using microRNA sequencing and histological analyses, we identified significantly elevated expression of miR487a-3p and miR6855-3p in plasma, peripheral monocytes, and coronary plaque macrophages of patients with CAD, which positively correlated with blood lipid levels and CAD severity. Through macrophage transcriptome sequencing and functional assays, we elucidated the molecular mechanisms by which miR487a-3p and miR6855-3p drive macrophage phenotypic switching and metabolic reprogramming

toward proinflammatory responses and excessive lipid retention, and identified CPE (carboxypeptidase E) and RRM2 (ribonucleotide reductase regulatory subunit M2) as their respective direct targets. Furthermore, using myeloid-specific CPE- and RRM2-deficient mice on a hypercholesterolemic background, we demonstrated that suppression of these targets accelerates atherosclerosis in vivo, thereby establishing a mechanistic link between the microRNA-target gene axis and atherosclerosis progression. These findings establish miR487a-3p and miR6855-3p as promising biomarkers for CAD diagnosis and prognosis and as potential therapeutic targets.

MATERIALS AND METHODS

Data Availability

All data supporting the findings of this study are included in the main text and the [Supplemental Material](#). All data, analytical methods, and materials are available from the corresponding author upon reasonable request. Detailed experimental materials and methods and a [Major Resources Table](#) are available in the [Supplemental Material](#).

Human Studies

After obtaining informed consent, venous blood was collected preoperatively from patients with CAD and non-CAD controls undergoing coronary angiography. Coronary arterial tissue was procured from patients with CAD undergoing coronary artery bypass graft surgery. Healthy coronary arterial tissue was obtained from organ donors. All subjects and donors were from Zhongnan Hospital of Wuhan University; clinical characteristics were presented in Tables 1, 2, 4, and 5 and [Tables S1, S2, and S4](#). All protocols for collecting and using human blood and coronary arterial samples were approved by the Medical Ethics Committee of Zhongnan Hospital of Wuhan University (protocol No. 2025058K). Blood samples were used for the isolation of plasma and immune cells, and subsequent RNA extraction and flow cytometry (immune cells), while coronary arterial samples were used for RNA extraction, as well as histological analysis.

Animal Studies

All animal experiments complied with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and were performed under Institutional Animal Care and Use Committee protocol no. ZN2023042 at Zhongnan Hospital of Wuhan University. Studies conformed to the Animal Research: Reporting of In Vivo Experiments 2.0 reporting guidelines. To eliminate potential confounding from the female estrous cycle,¹⁶ male mice at 7 weeks old were used for the experiments.

Assessment of Atherosclerotic Burden

Atherosclerotic lesions were evaluated in en face preparations of the whole mouse aortas, as well as in cross-sections of the mouse aortic roots and human coronary arteries. The detailed experimental protocol is provided in the [Supplemental Material](#).

Table 1. Demographic Data of the HCs and Patients With CAD for Monocyte miR-seq

	HC (n=6)	CAD (n=7)	P value
Age, y	62.50±4.51	64.00±4.20	0.5474
Male, n (%)	3 (50.00%)	4 (57.14%)	–
Body mass index, kg/m ²	21.78±0.95	24.22±3.02	0.0811
Smoking, n (%)	0 (0%)	4 (57.14%)	0.0699
Drinking, n (%)	0 (0%)	4 (57.14%)	0.0699
Hypertension, n (%)	0 (0%)	0 (0%)	–
Diabetes, n (%)	0 (0%)	0 (0%)	–
Statin use, n (%)	4 (66.67%)	7 (100.00%)	0.1923
ACE inhibitor/ARB use, n (%)	0 (0%)	3 (42.86%)	0.1923
β-Blocker use, n (%)	2 (33.33%)	2 (28.57%)	–
White blood cells (×10 ⁹ /L)	5.46 (5.38–7.64)	6.47 (5.40–7.91)	0.6562
Neutrophils (×10 ⁹ /L)	3.92±1.25	4.23±1.30	0.6702
Lymphocytes (×10 ⁹ /L)	1.98±0.69	1.75±0.53	0.524
Monocytes (×10 ⁹ /L)	0.45±0.09	0.46±0.07	0.7184
Alanine aminotransferase, U/L	10.00 (9.25–25.75)	17.00 (9.00–23.00)	0.7587
Aspartate aminotransferase, U/L	17.67±3.39	20.29±6.26	0.3811
Total bilirubin, μmol/L	11.30±4.21	12.33±3.51	0.6401
Blood urea nitrogen, mmol/L	6.07±0.90	6.11±0.61	0.9373
Creatinine, μmol/L	64.23±12.44	71.66±12.89	0.3155
Total cholesterol, mmol/L	4.32±0.63	5.30±0.69	0.0229
Triglycerides, mmol/L	0.98 (0.61–1.60)	1.51 (0.86–1.67)	0.1678
HDL-C, mmol/L	1.44±0.33	1.31±0.29	0.4547
LDL-C, mmol/L	2.13±0.60	2.83±0.44	0.0346
sdLDL-C, mmol/L	0.67±0.23	1.11±0.36	0.0236
Fasting blood glucose, mmol/L	5.00±0.33	5.68±0.81	0.081
Left ventricular ejection fraction, %	66.17±1.60	62.57±5.13	0.1202

Data are expressed as mean±SD or median with interquartile range in nonnormal distribution and n (%). – indicates $P>0.9999$; ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CAD, coronary artery disease; HC, health control; HDL-C, high-density lipoprotein-cholesterol; LDL-C, low-density lipoprotein-cholesterol; miR-seq, microRNA sequencing; and sdLDL-C, small dense low-density lipoprotein-cholesterol.

MicroRNA-Sequencing Analyses

Total RNA was isolated from peripheral blood monocytes of healthy controls (HCs) and patients with CAD using TRIzol reagent (Invitrogen, 15596026). The clinical characteristics of all subjects are listed in Table 1. MicroRNA libraries were generated using NEB Next Multiplex Small RNA Library Prep Set for Illumina (NEB, E7300L). Library quality control and quantification were assessed using the Agilent 5400 Fragment Analyzer System and quantitative real-time polymerase chain reaction (1.5 nM), respectively. Qualified libraries with insertions between 18 and 40 bp were sequenced on the Illumina NovaSeq 6000 platform (Novogene, Inc, Beijing, China) with the SE50 strategy. Raw data from sequencing were base-called using Illumina Casava software (version 2.19) and filtered by Fastp (version 0.23.2).¹⁷ Clean reads were obtained by removing reads containing poly-N or 5' adapter contaminants, reads lacking 3' adapter or the inserted tag, and reads containing poly A or T or G or C and low-quality reads. Simultaneously, Q20, Q30, and GC content of the raw data were calculated. The quality control of sequencing reads was performed with FastQC (version 0.12.1).¹⁸ The clean reads were aligned to Ensembl human genome, Genome Reference Consortium human build 38¹⁸ and a custom sequence set¹⁹

containing miRBase (version 20.0) known human microRNA hairpins, tRNA, rRNA, adapter, and predicted novel microRNA hairpin sequences using Bowtie (version 0.12.9).²⁰ miRDeep2 (version 2.0.0.8)²¹ was used to count reads mapped to the microRNA database. miRevo (version 1.1)²² and miRDeep2^{19,21} were integrated to predict novel microRNAs. Target gene prediction was performed using the TargetScan (version 8.0) database.¹⁹ MicroRNA expression levels were estimated by transcript per million using established criteria.²³ Differential expression analysis of 2 groups was performed using the differential expression analysis for sequence data 2 Bioconductor R package (version 1.12.0).¹⁸ The P values were adjusted using the Benjamini and Hochberg method, with an adjusted P value of 0.05 set as the threshold for significant differential expression. MicroRNA-seq data have been deposited in the Gene Expression Omnibus database under accession number GSE335640.

Gene Set Enrichment Analysis (GSEA) was performed on the predicted target genes of differentially expressed microRNAs using the Gene Ontology (GO) analyzer for RNA sequencing (RNA-seq) and other length-biased data (version 1.64.0),²⁴ which accounts for gene length bias. Statistical enrichment of target gene candidates in Kyoto Encyclopedia of Genes

Table 2. Demographic Data of the Ctrls and Patients With CAD for Plasma miRNA Expression Assay

	Ctrl (n=43)	CAD (n=59)	P value
Age, y	61.12±6.38	63.73±9.13	0.0921
Male, n (%)	25 (58.14%)	44 (74.58%)	0.0905
Body mass index, kg/m ²	23.44 (21.48–25.39)	24.80 (22.72–26.22)	0.1499
Smoking, n (%)	9 (20.93%)	18 (30.51%)	0.3646
Drinking, n (%)	5 (11.63%)	13 (22.03%)	0.199
Hypertension, n (%)	7 (16.28%)	19 (32.20%)	0.1063
Diabetes, n (%)	3 (6.98%)	11 (18.64%)	0.144
Statin use, n (%)	36 (83.72%)	59 (100%)	0.0017
ACE inhibitor/ARB use, n (%)	2 (4.65%)	29 (49.15%)	<0.0001
β-Blocker use, n (%)	5 (11.63%)	30 (50.85%)	<0.0001
White blood cells (×10 ⁹ /L)	5.91 (5.30–6.60)	6.07 (5.10–7.20)	0.6597
Neutrophils (×10 ⁹ /L)	3.61 (3.20–4.12)	3.80 (3.20–4.80)	0.4773
Lymphocytes (×10 ⁹ /L)	1.67±0.51	1.55±0.44	0.2059
Monocytes (×10 ⁹ /L)	0.45 (0.40–0.51)	0.50 (0.40–0.60)	0.1558
Alanine aminotransferase, U/L	17.00 (11.00–27.00)	19.00 (14.00–26.00)	0.2767
Aspartate aminotransferase, U/L	18.00 (16.00–23.00)	19.00 (17.00–24.00)	0.3269
Total bilirubin, μmol/L	12.80 (10.00–16.20)	11.70 (8.30–14.30)	0.1712
Blood urea nitrogen, mmol/L	5.50 (4.80–7.07)	6.40 (5.20–7.50)	0.1096
Creatinine, μmol/L	67.80 (58.50–86.30)	73.60 (66.00–84.90)	0.1578
Total cholesterol, mmol/L	4.12 (3.53–4.75)	5.20 (4.20–5.42)	<0.0001
Triglycerides, mmol/L	1.14 (0.90–1.55)	1.40 (1.13–1.61)	0.0658
HDL-C, mmol/L	1.09 (0.96–1.35)	1.09 (0.93–1.24)	0.6402
LDL-C, mmol/L	2.36 (1.85–2.77)	3.01 (2.53–3.37)	<0.0001
sdLDL-C, mmol/L	0.73 (0.49–0.96)	1.00 (0.79–1.23)	0.0004
Fasting blood glucose, mmol/L	5.32 (5.00–5.63)	5.52 (4.90–6.43)	0.2803
Left ventricular ejection fraction, %	65.00 (62.00–67.00)	64.00 (60.00–68.00)	0.3943

Data are expressed as mean±SD or median with interquartile range in nonnormal distribution and n (%). ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CAD, coronary artery disease; Ctrl, control; HDL-C, high-density lipoprotein-cholesterol; LDL-C, low-density lipoprotein-cholesterol; miRNA, microRNA; and sdLDL-C, small dense low-density lipoprotein-cholesterol.

and Genomes (KEGG) pathways was evaluated using KEGG Orthology Based Annotation System software (version 3.0).²⁵

mRNA-Sequencing Analyses

Total RNA was extracted from THP-1 (Tohoku hospital pediatrics-1)-derived macrophages with the indicated treatment or from peripheral monocytes of HCs and patients with CAD using TRIzol reagent. The clinical information of all subjects is listed in Table 5. mRNA was purified from total RNA using poly T

oligo-attached magnetic beads. RNA libraries for THP-1-derived macrophages were prepared using Fast RNA-seq Lib Prep Kit V2 (ABclonal, RK20306), and RNA-seq was performed on an Illumina NovaSeq X Plus platform (Novogene). RNA libraries for peripheral monocytes were prepared using NEBNext Ultra Directional RNA Library Prep Kit for Illumina (NEB, E7420), and RNA-seq was performed on an Illumina NovaSeq 6000 platform (Novogene). Library quality control and quantification were performed with Agilent Bioanalyzer 2100 and quantitative real-time polymerase chain reaction, respectively. Raw sequencing

Table 3. Multivariable Logistic Regression Analysis of Plasma miRNA Levels and Lipid Parameters Associated With Coronary Artery Disease

Variables	B	SE	Wald	P value	OR	95% CI
miR487a-3p	0.86	0.37	5.40	0.0202	2.36	1.14–4.86
miR6855-3p	1.90	0.79	5.80	0.0161	6.72	1.43–31.67
TC	0.28	0.51	0.31	0.5772	1.33	0.49–3.57
LDL-C	0.60	0.65	0.87	0.3511	1.82	0.52–6.46
sdLDL-C	0.26	0.83	0.09	0.7587	1.29	0.25–6.59

B indicates unstandardized coefficient; LDL-C, low-density lipoprotein-cholesterol; miRNA, microRNA; OR, odds ratio; sdLDL-C, small dense low-density lipoprotein-cholesterol; TC, total cholesterol; and Wald, Wald χ^2 statistic.

Table 4. Demographic Data of the Ctrls and Patients With CAD for miRNA Expression Assay in Coronary Arteries

	Ctrl (n=10)	CAD (n=18)	P value
Age, y	57.60±5.54	58.39±7.97	0.7840
Male, n (%)	8 (80.00%)	14 (77.78%)	–
Body mass index, kg/m ²	23.74±2.02	22.64±2.44	0.2379
Smoking, n (%)	0 (0%)	1 (5.56%)	–
Drinking, n (%)	0 (0%)	0 (0%)	–
Hypertension, n (%)	2 (20.00%)	8 (44.44%)	0.2474
Diabetes, n (%)	0 (0%)	3 (16.67%)	0.5330
Statin use, n (%)	0 (0%)	18 (100%)	<0.0001
ACE inhibitor/ARB use, n (%)	0 (0%)	1 (5.56%)	–
β-Blocker use, n (%)	1 (10.00%)	7(38.89%)	0.1937
Total cholesterol, mmol/L	N/A	5.42±0.37	N/A
Triglycerides, mmol/L	N/A	1.60±0.51	N/A
HDL-C, mmol/L	N/A	1.12±0.28	N/A
LDL-C, mmol/L	N/A	3.43±0.42	N/A
sdLDL-C, mmol/L	N/A	1.12±0.24	N/A

Data are expressed as mean±SD and n (%). For organ donor Ctrl, part of the clinical data was N/A. – indicates $P>0.9999$; ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CAD, coronary artery disease; Ctrl, control; HDL-C, high-density lipoprotein-cholesterol; LDL-C, low-density lipoprotein-cholesterol; miRNA, microRNA; N/A, not available; and sdLDL-C, small dense low-density lipoprotein-cholesterol.

data were processed using Illumina Casava (version 2.19) and filtered by Fastp (version 0.23.2).¹⁷ Clean reads were obtained by removing adapter-containing reads, ploy-N reads, and low-quality reads, while Q20, Q30, and GC content of the clean data were calculated. Sequencing read quality was assessed using FastQC (version 0.12.1).¹⁸ HISAT2 software (version 2.2.1)¹⁸ was used to align clean reads to the Ensembl human genome, Genome Reference Consortium human build 38.¹⁸ The mapped reads of each sample were assembled by StringTie (version 2.2.3)¹⁸ in a reference-based approach. FeatureCounts (version 2.0.6)²⁶ was used to count reads mapped to each gene, and fragments per kilobase of transcript sequence per million base pairs sequenced of each gene were calculated based on the gene length and mapped reads. Differential expression analysis for 2 groups was performed using the differential expression analysis for sequence data 2 (version 1.42.0).¹⁸ P values were adjusted using the Benjamini-Hochberg method to control the false discovery rate. The threshold for significant differential expression was set at adjusted $P\leq 0.05$ and $|\log_2(\text{fold change})|\geq 1$. RNA-seq data were deposited into the Gene Expression Omnibus database under accession numbers GSE335633 (THP-1-derived macrophages) and GSE336634 (peripheral monocytes).

GSEA was performed using the Java GSEA desktop application (Broad Institute). Gene sets were defined according to KEGG pathways and GO terms. For each gene set, enrichment was evaluated using a ranked list of genes and gene set permutation. Gene sets with a false discovery rate $<0.25^{27}$ were considered statistically significant.

Data Analysis

Data are expressed as mean±SD or median with interquartile range in a nonnormal distribution. Normality was assessed

using the Shapiro-Wilk test, and group variances were compared using the F test for data sets with 2 groups or the Brown-Forsythe test with groups of ≥ 3 . For normally distributed data, the unpaired Student t test with equal or unequal variance (Welch correction) was used for 2-group analyses, and a 1- or 2-way ANOVA test was used for multigroup analyses. For nonnormal distributions, the Mann-Whitney U test was used for 2-group analyses. Categorical data were analyzed using the Fisher exact test and presented as counts (percentages). The Pearson or Spearman (for nonnormal distributions) correlation analysis was used to assess the correlation between the 2 indicators. Univariable and multivariable logistic regression analyses were performed to evaluate the independent association of plasma microRNA levels and lipid parameters with CAD status. Specific approaches are detailed in the figure legends. All statistical analyses were performed with GraphPad Prism 10.4.0 (GraphPad Software). A P value of <0.05 was considered statistically significant.

RESULTS

miR487a-3p and miR6855-3p Are Upregulated in Monocytes and Plasma of Patients With CAD and Correlate With Atherosclerosis Severity

To identify microRNAs dysregulated in CAD, we performed microRNA sequencing on peripheral monocytes from patients with CAD and HCs (Figure 1A). The purity of negatively selected monocytes from peripheral blood mononuclear cells of nonhypertensive and nondiabetic subjects reached $\approx 90\%$ (Figure 1B). Blood lipid levels, including total cholesterol (TC), LDL-C (low-density lipoprotein-cholesterol), and sdLDL-C (small dense LDL-C), were significantly higher in patients with CAD compared with HCs, whereas other biochemical parameters, hematology, and left ventricular ejection fraction were comparable between groups (Table 1). A total of 50 microRNAs were differentially expressed in monocytes from patients with CAD versus HCs (Figure 1C). A heatmap of the top 17 microRNAs ($|\log_2[\text{fold change}]|\geq 1.9$) revealed 6 markedly upregulated and 11 downregulated microRNAs (Figure 1D). GO and KEGG enrichment analyses of the target genes of these 50 microRNAs revealed pathways related to inflammation, immune activation, and lipid metabolism (Figure S1A and S1B).

To validate these findings, we measured the expression of the top 17 microRNAs in an expanded cohort of peripheral monocytes and peripheral blood mononuclear cells from HCs and patients with CAD (Tables S1 and S2). Among these, miR6855-3p, miR765, and miR487a-3p were consistently upregulated in both peripheral blood mononuclear cells and monocytes from patients with CAD compared with HCs, confirming the microRNA sequencing results (Figure 1E; Figure S2A and S2B).

Given the hyperlipidemic profile of patients with CAD, we next tested whether these 3 microRNAs were responsive to lipid stimulation. In both human monocyte-derived

Table 5. Demographic Data of the HCs and Patients With CAD for Monocyte mRNA Sequencing

	HC (n=10)	CAD (n=9)	P value
Age, y	60.70±5.98	64.00±5.79	0.2395
Male, n (%)	5 (50.00%)	5 (55.56%)	–
Body mass index, kg/m ²	22.35±2.13	24.04±2.73	0.1494
Smoking, n (%)	1 (10.00%)	5 (55.56%)	0.0573
Drinking, n (%)	1 (10.00%)	4 (44.44%)	0.1409
Hypertension, n (%)	0 (0%)	0 (0%)	–
Diabetes, n (%)	0 (0%)	0 (0%)	–
Statin use, n (%)	7 (70.00%)	9 (100.00%)	0.2105
ACE inhibitor/ARB use, n (%)	0 (0%)	3 (33.33%)	0.0867
β-Blocker use, n (%)	2 (20.00%)	3 (33.33%)	0.6285
White blood cells (×10 ⁹ /L)	5.46 (5.30–7.25)	6.56 (5.90–7.81)	0.2342
Neutrophils (×10 ⁹ /L)	3.86±1.15	4.37±1.17	0.3534
Lymphocytes (×10 ⁹ /L)	1.84±0.55	1.68±0.49	0.539
Monocytes (×10 ⁹ /L)	0.46±0.09	0.49±0.08	0.4693
Alanine aminotransferase, U/L	11.00 (10.00–25.00)	17.00 (10.00–22.00)	0.9222
Aspartate aminotransferase, U/L	17.60±3.31	20.11±5.44	0.2352
Total bilirubin, μmol/L	12.03±4.19	11.88±3.37	0.932
Blood urea nitrogen, mmol/L	6.23±0.81	6.02±0.63	0.5305
Creatinine, μmol/L	68.41±13.77	73.02±11.60	0.4436
Total cholesterol, mmol/L	4.28±0.55	5.26±0.60	0.0017
Triglycerides, mmol/L	0.81 (0.61–1.26)	1.40 (0.89–1.60)	0.0751
HDL-C, mmol/L	1.45±0.33	1.27±0.31	0.2542
LDL-C, mmol/L	2.18±0.60	2.93±0.55	0.012
sdLDL-C, mmol/L	0.66±0.22	1.14±0.32	0.0014
Fasting blood glucose, mmol/L	5.20±0.37	5.50±0.78	0.3065
Left ventricular ejection fraction, %	65.70±2.79	63.56±4.95	0.2546

Data are expressed as mean±SD or median with interquartile range in nonnormal distribution and n (%). – indicates $P>0.9999$; ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CAD, coronary artery disease; HC, health control; HDL-C, high-density lipoprotein-cholesterol; LDL-C, low-density lipoprotein-cholesterol; and sdLDL-C, small dense low-density lipoprotein-cholesterol.

macrophages (hMDMs) and THP-1-derived macrophages, ox-LDL (oxidized LDL) robustly induced the expression of miR487a-3p and miR6855-3p, but not miR765, over 48 hours (Figure 1F). This lipid-inducible expression pattern led us to focus on miR487a-3p and miR6855-3p for further investigation, excluding miR765. To identify the upstream regulators mediating this induction, we used the TransmiR database to screen candidate TFs (transcription factors). Among 35 candidate TFs for miR487a-3p, only 2 were predicted to activate its transcription (Figure S3A), and a ChIP (chromatin immunoprecipitation) assay confirmed the binding of KLF (Krüppel-like factor) 5 to the miR487a-3p promoter (Figure S3C). The ox-LDL markedly induced KLF5 expression in THP-1-derived macrophages, and KLF5 silencing significantly reduced miR487a-3p expression both at baseline and under ox-LDL stimulation (Figure 1G; Figure S3B), establishing KLF5 as the mediator of ox-LDL-induced miR487a-3p transcription. For miR6855-3p, parallel bioinformatic screening and gene expression

validation identified IRF1 (interferon regulatory factor 1) and EGR1 (early growth response 1) as candidate TFs (Figure S3D), both of which were activated by ox-LDL (Figure 1H; Figure S3E and S3F). Silencing of IRF1, but not EGR1, significantly reduced miR6855-3p expression under both basal and ox-LDL-stimulated conditions (Figure 1H; Figure S3F), identifying IRF1 as the primary TF driving ox-LDL-induced miR6855-3p transcription, with the predicted binding site shown in Figure S3H.

Given that cellular microRNAs can be released into the extracellular environment and are detectable in biofluids, including plasma, follicular fluid, and even in cell culture media,^{28,29} we tested whether miR487a-3p and miR6855-3p were secreted from macrophages. Both microRNAs were robustly released into the supernatant of THP-1-derived macrophages and hMDMs upon ox-LDL treatment, paralleling their intracellular upregulation (Figure S4A and S4B). Moreover, silencing of KLF5 or IRF1 significantly reduced their extracellular release (Figure S4A), indicating that intracellular upregulation in

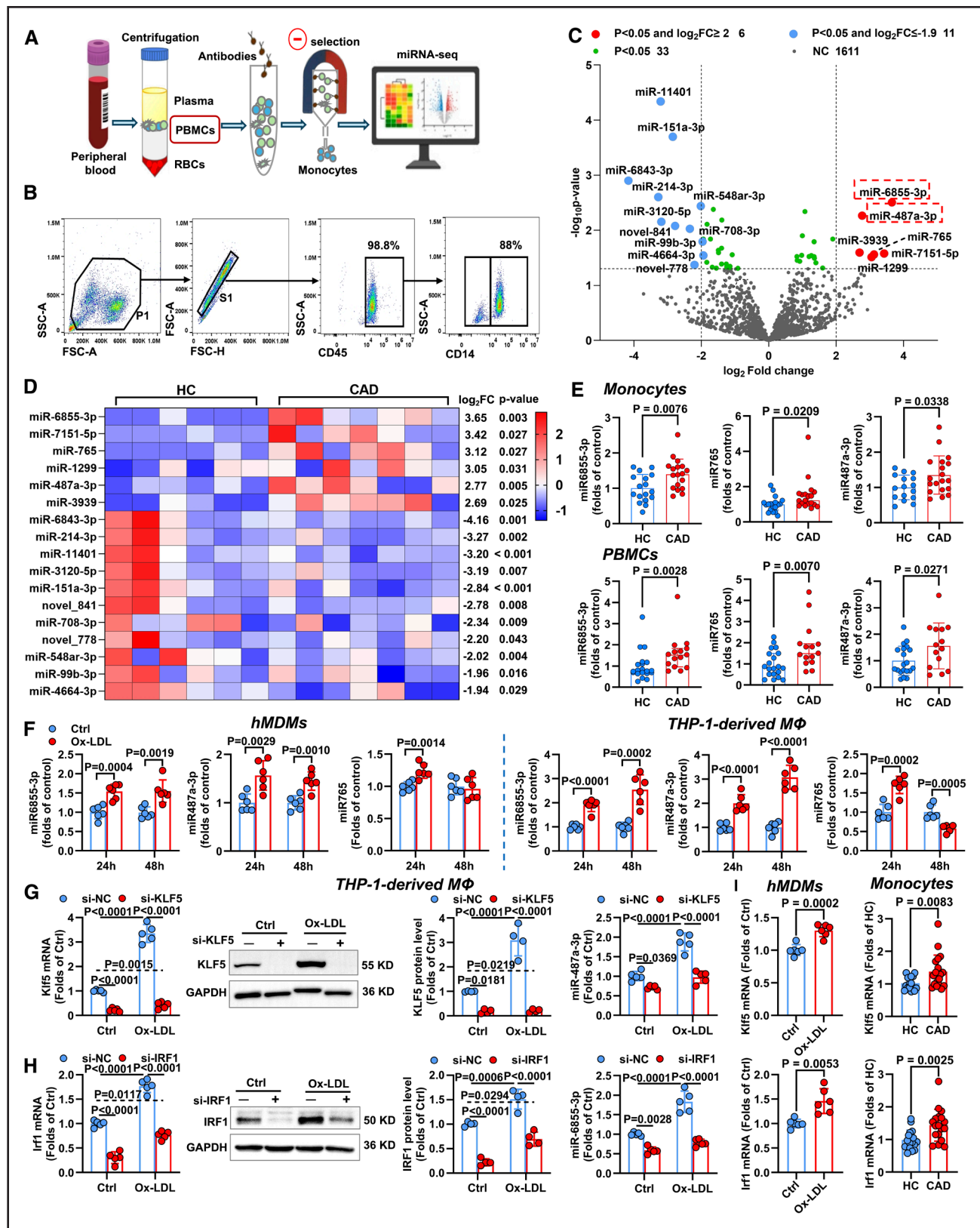


Figure 1. miR487a-3p and miR6855-3p are increased in peripheral monocytes of patients with coronary artery disease (CAD) compared with healthy controls (HCs).

A, Schematic showing the process of peripheral blood mononuclear cells (PBMCs) collection, monocyte isolation, and microRNA sequencing (miR-seq) on Illumina platforms. **B**, Representative flow cytometry plots showing peripheral leukocytes (CD45⁺) and monocytes (CD14⁺). **C**, Volcano plot displaying differentially expressed miRNAs in peripheral monocytes from HCs (n=6) and patients with CAD (n=7). (Continued)

macrophages contributes to their extracellular levels. We, therefore, measured plasma miR487a-3p and miR6855-3p levels in a large cohort of patients with CAD and controls who had comparable demographic characteristics but differed in blood lipid levels and medication history (Table 2; more anti-hypertensive and cholesterol-lowering drug use in CAD versus control). Plasma levels of both microRNAs were significantly elevated in patients with CAD compared with controls and positively correlated with blood TC, LDL-C, and sdLDL-C but not with triglycerides, high-sensitivity C-reactive protein, blood pressure, or fasting blood glucose (Figure 2A; Figure S4C and S4D). Both microRNAs were comparable between smoking and nonsmoking patients with CAD (Figure S4E). Given that sdLDL-C is the most atherogenic lipoprotein parameter, and both sdLDL-C and LDL-C contribute significantly to atherosclerotic cardiovascular disease,^{30,31} these results suggest a link between elevated circulating miR487a-3p and miR6855-3p and atherosclerosis in patients with CAD.

To further evaluate the clinical relevance of these circulating microRNAs, we examined their relationship with coronary atherosclerosis severity using the Gensini score, a quantitative measure of total coronary atherosclerotic burden that provides valuable information on CAD severity and prognosis.^{32,33} Plasma miR487a-3p and miR6855-3p levels positively correlated with the Gensini score (Figure 2A). Receiver operating characteristic (ROC) analysis was performed to assess their diagnostic performance. The area under the ROC curve for miR487a-3p was 0.841, with a sensitivity of 76.27% and a specificity of 76.74% (cutoff value, 1.175), while the ROC curve for miR6855-3p was 0.829, with a sensitivity of 80.33% and a specificity of 74.42% (cutoff value, 1.155; Figure 2A). These results demonstrate the reliability of circulating miR487a-3p and miR6855-3p as diagnostic biomarkers for CAD. To assess whether miR487a-3p and miR6855-3p are independent predictors of CAD beyond their association with lipid profiles, we performed single-factor and multifactor logistic regression analyses. In univariate analysis, miR487a-3p (odds ratio [OR], 4.322; $P=0.0001$), miR6855-3p (OR, 22.258; $P<0.0001$), TC (OR, 2.89; $P=0.0001$), LDL-C (OR, 3.417; $P=0.0003$), and sdLDL-C (OR, 4.935; $P=0.0061$) were all significantly associated with CAD (Table S3). In multivariate logistic regression incorporating all 3 lipid

parameters, both microRNAs remained significant independent predictors: miR487a-3p (OR, 2.356; $P=0.0202$) and miR6855-3p (OR, 6.718; $P=0.0161$), whereas none of the lipid parameters retained significance (Table 3). These data demonstrate that plasma miR487a-3p and miR6855-3p are independent risk factors for CAD, and their predictive value is not merely a reflection of circulating lipid levels.

miR487a-3p and miR6855-3p Are Enriched in Coronary Plaque Macrophages and Correlate With Lesion Area

To further understand the translational relevance of these findings, we examined the expression of miR487a-3p and miR6855-3p in coronary atherosclerotic plaques, where macrophages derived from infiltrating monocytes play a central role.^{10,11,34,35} Coronary arterial tissue was procured from patients with severe CAD undergoing coronary artery bypass graft and from control donor hearts unsuitable for transplantation (Table 4). Oil Red O staining revealed that the plaque area occupied $\approx 72\%$ of the coronary arterial wall in the CAD group compared with 11% in controls (Figure 2B). Accordingly, miR487a-3p and miR6855-3p expressions were dramatically increased in the coronary arteries of patients with CAD versus controls (Figure 2C). Consistent with previous reports,^{8,11,35} macrophages (CD68⁺) were significantly expanded (≈ 10 -fold) in the arterial wall of patients with CAD compared with controls (Figure 2D; Figure S4F).

We next determined the cellular source of these microRNAs using fluorescence in situ hybridization combined with immunofluorescence staining for CD68. Macrophages constituted $\approx 93\%$ of coronary arterial miR487a-3p⁺ or miR6855-3p⁺ cells in both groups. However, the proportion of macrophages expressing these microRNAs differed markedly: in patients with CAD, 66.1% of macrophages were miR487a-3p⁺, and 57.7% were miR6855-3p⁺, compared with only 8.3% and 8.4% in controls, respectively (Figure 2D). Consequently, miR487a-3p⁺ and miR6855-3p⁺ macrophages were massively expanded (>30 -fold versus controls) in the coronary arterial wall of patients with CAD, accounting for the elevated expression of both microRNAs in CAD coronary tissue. In contrast, miR487a-3p⁺ and miR6855-3p⁺ macrophages were barely detectable in

Figure 1 Continued. **D**, Heatmap showing expression levels of the 17 differentially expressed miRNAs (red and blue dots in **C**). **E**, The expressions of miR6855-3p, miR487a-3p, and miR765 in peripheral monocytes (HC: 18 and CAD: 19) and PBMCs (HC: 20 and CAD: 15) from HCs and patients with CAD. **F**, The expressions of miR6855-3p, miR487a-3p, and miR765 in human monocyte-derived macrophages (hMDMs; donor: 6) and THP-1 (Tohoku hospital pediatrics-1)-derived macrophages treated with vehicle (control [Ctrl]) or 100- μ g/mL ox-LDL (oxidized low-density lipoprotein) for 24 or 48 hours ($n=6$). The mRNA and protein levels of KLF5 (Krüppel-like factor 5; **G**) or IRF1 (interferon regulatory factor 1; **H**) and the expressions of miR487a-3p (**G**) or miR6855-3p (**H**) in THP-1-derived macrophages transfected with negative control (NC) or KLF5 (**G**) or IRF1 (**H**) siRNA, followed by treatment with Ctrl or 100- μ g/mL ox-LDL for 24 hours. **I**, mRNA expressions of KLF5 and IRF1 in hMDMs (donor: 6) treated with Ctrl or ox-LDL for 24 hours or in peripheral monocytes from HCs ($n=17$) and patients with CAD ($n=19$). Statistics: the unpaired Student *t* test in **E**, **F**, and **I**, and the Mann-Whitney *U* test in **E** and **I**. Two-way ANOVA with the post hoc Tukey multiple comparisons between groups in **G** and **H**.

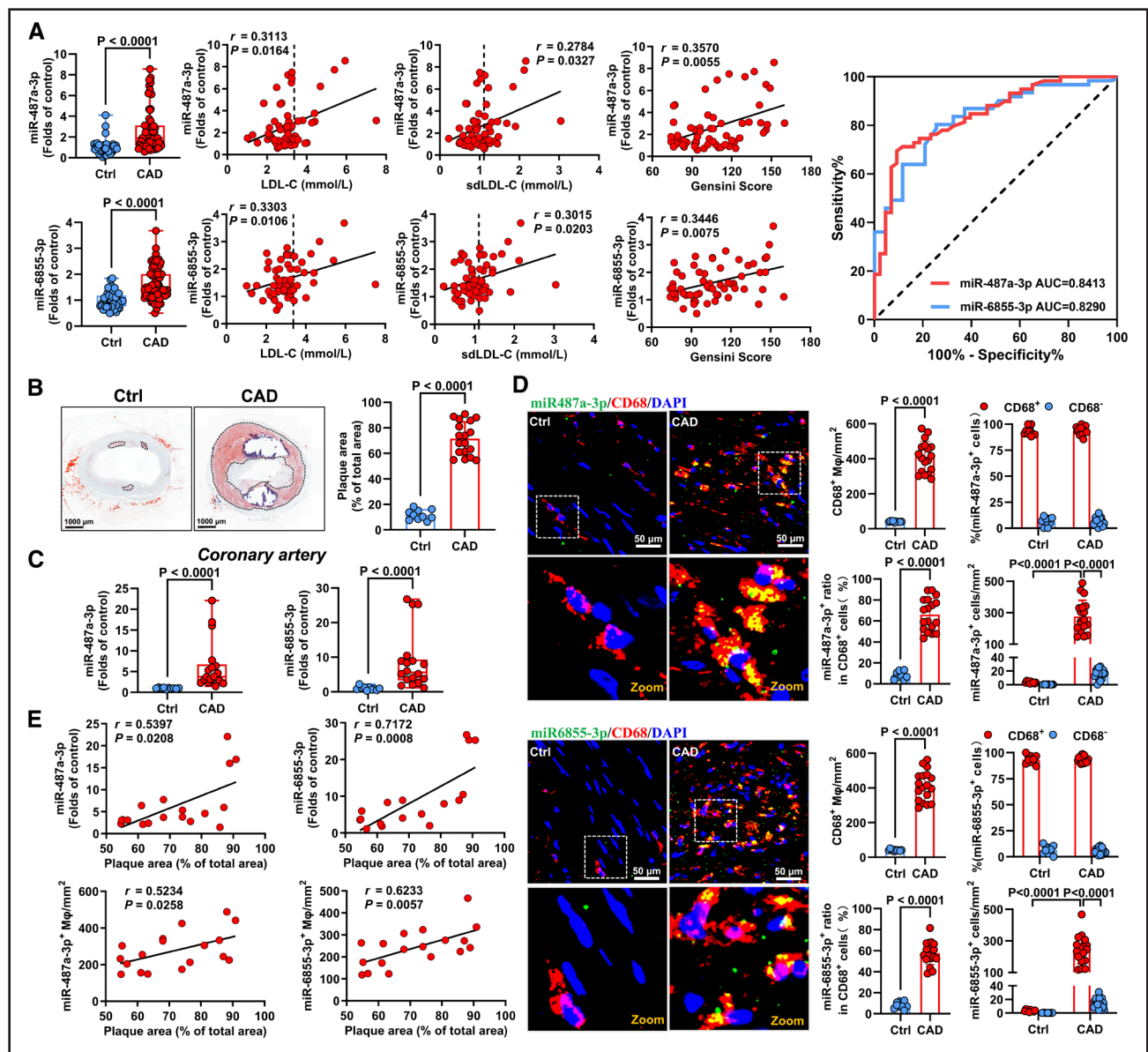


Figure 2. Increases of miR487a-3p and miR6855-3p in plasma and coronary arteries of patients with coronary artery disease (CAD) correlate with hyperlipidemia and plaque area, respectively.

A, Plasma miR487a-3p and miR6855-3p levels in controls (Ctrls) and patients with CAD (left). Correlations between plasma miR487a-3p or miR6855-3p and LDL-C (low-density lipoprotein-cholesterol), sdLDL-C (small dense LDL-C), and Gensini score in patients with CAD (middle). Receiver operating characteristic (ROC) curves for evaluating the diagnostic value of miR487a-3p and miR6855-3p for CAD (right). Ctrl, n=43; CAD, n=59. **B**, Representative Oil Red O staining of coronary artery sections from Ctrls and patients with CAD, with corresponding quantitation of plaque area ratios. Plaque areas were demarcated by dashed outlines. Scale bar, 1000 μ m. **C**, The expressions of miR487a-3p and miR6855-3p in coronary arteries of Ctrls and patients with CAD. **D**, Representative low- and high-magnification confocal images of coronary arteries from Ctrls and patients with CAD stained for CD68 (red) and fluorescent miR487a-3p and miR6855-3p in situ hybridization (green), with quantitation for the numbers and frequencies of CD68⁺, miR487a-3p⁺, and miR6855-3p⁺ cells. The 4',6-diamidino-2-phenylindole (DAPI; blue) was used to label nuclei. Scale bar, 50 μ m. Ctrls (n=10) and CAD (n=18) in B through D. **E**, Correlations between plaque area ratios and coronary artery miR487a-3p or miR6855-3p levels, and miR487a-3p⁺ or miR6855-3p⁺ macrophage abundance in patients with CAD (n=18). Statistics: the unpaired Student *t* test in D, the Welch *t* test in B and D, and the Mann-Whitney *U* test in A, C, and D. Pearson correlation analysis in E and Spearman correlation analysis in A and E.

control arteries, where few macrophages were present, and both microRNAs were expressed at low levels (Figure 2D; Figure S4F).

Notably, coronary plaque area in patients with CAD was positively correlated with both arterial miR487a-3p

and miR6855-3p expression and with the abundance of miR487a-3p⁺ and miR6855-3p⁺ macrophages in the arterial wall (Figure 2E). Together, these results demonstrate that miR487a-3p and miR6855-3p are predominantly expressed in arterial macrophages, miR487a3p⁺

and miR6855-3p⁺ macrophages expand robustly in atherosclerotic plaques, and their abundance is associated with lesion burden, implicating macrophage-derived miR487a-3p and miR6855-3p in atherogenesis.

miR487a-3p and miR6855-3p Reprogram Macrophages Toward Proinflammatory and Proatherogenic Gene Expression

To investigate how miR487a-3p and miR6855-3p modulate macrophage function in atherogenesis, we performed mRNA sequencing on THP-1-derived macrophages transfected with miR487a-3p mimic, miR6855-3p mimic, or negative control (NC) for 24 hours (Figure S5A and S6A).

miR487a-3p overexpression induced 2852 gene upregulation and 3112 gene downregulation versus NC (Figure 3A). GSEA-based GO analysis revealed enrichment of terms related to proinflammatory responses, including positive regulation of the MAPK (mitogen-activated protein kinase) cascade, NF- κ B (nuclear factor- κ B) signaling, cytokine release and binding, chemotaxis, and immune cell activation, as well as proatherogenic processes such as apoptosis, response to lipid, and suppression of lipid storage and transport and extracellular matrix organization (Figure 3B). GSEA-based KEGG analysis further demonstrated upregulation of pathways involved in lipid metabolism and atherosclerosis, chemokine and cytokine-cytokine receptor interactions, and inflammatory signaling pathways, including TLR (Toll-like receptor), TNF (tumor necrosis factor), NOD (nucleotide-binding oligomerization domain)-like receptor, JAK/STAT (Janus kinase/signal transduction and transcription activation), NF- κ B, and MAPK (Figure 3C). Heatmaps displaying DEGs by functional category (Figure 3D and 3E; Figure S5B and S5C) were validated by quantitative reverse transcription polymerase chain reaction (Figure S5D through S5F). C-C and C-X-C motif chemokine ligands (CCLs [C-C motif chemokine ligands] and CXCLs [C-X-C motif chemokine ligands]) and their receptors (CCRs [C-C chemokine receptors] and CXCRs [C-X-C motif chemokine receptors]) orchestrate monocyte and macrophage mobilization, recruitment, and fate during atherosclerosis initiation and progression.^{36–38} miR487a-3p overexpression upregulated cytokines, such as CSF (colony-stimulating factor) 1, CSF2, CSF3, IL (interleukin)-1 α , IL-15, IL-27, and TNFSF (TNF superfamily) 10, chemokines (CCL3, CCL3L1, CCL4, CCL5, CCL8, CCL20, CXCL1, CXCL2, CXCL3, CXCL8, CXCL9, CXCL10, CXCL11, CXCL12, and CXCL13), and their receptors (CCR1, CCR4, CCR7, CCRL2, CXCR4, and CXCR5; Figure 3D; Figure S5D). Proinflammatory polarization markers, including IL-6, CCL4L2, CD80, TNF, IL-1B, IFN (interferon) b1, and CCL2, were upregulated, whereas anti-inflammatory markers, such as CD206 (Mrc1 [mannose receptor C type 1]), CD163, VEGF (vascular endothelial growth factor) b, and IL-4R, were downregulated (Figure 3E;

Figure S5E). Furthermore, cholesterol efflux genes (*APOE* [apolipoprotein E], *ABCA1* [ATP binding cassette subfamily A member 1], *ABCG1* [ATP binding cassette subfamily G member 1], *SCARB1* [scavenger receptor class B member] 1, and *SCARB2*) were suppressed, while the cholesterol uptake receptor OLR1 (ox-LDL receptor 1) was increased, with only a mild decrease in CD36 (Figure 3E; Figure S5E), indicating a net shift toward lipid accumulation. In addition, miR487a-3p perturbed extracellular matrix homeostasis by downregulating TIMP (tissue inhibitor of matrix metalloproteinase) 1, TIMP2, TIMP3, MMP (matrix metalloproteinase) 2, MMP7, MMP9, MMP14, and MMP17 and upregulating MMP 3, MMP8, MMP10, and MMP13 (Figure S5B). Key TFs were also altered, with upregulation of KLF2, KLF4, KLF5, and NF κ B1, and downregulation of SREBF (sterol regulatory element binding transcription factor) 1, SREBF2, PPAR (peroxisome proliferator-activated receptor) A, PPARG, RHOA (ras homolog family member A), and RXR (retinoid x receptor) A (Figure S5C and S5F).

Similarly, miR6855-3p overexpression induced 1533 gene upregulation and 1453 gene downregulation events versus NC (Figure 4A). GSEA-based GO and KEGG analyses revealed enrichment of genes associated with enhanced inflammation, immune dysregulation, and abnormal lipid metabolism (Figure 4B and 4C). miR6855-3p overexpression upregulated cytokines (IL-15, IL-27, CSF1, TNFSF10, and TGF [transforming growth factor] B1), chemokines (CCL3, CCL3L1, CCL4, CCL7, CCL8, CXCL10, CXCL11, and CXCL12), and their receptors (CCR1 and CCR7; Figure 4D; Figure S6D). Proinflammatory polarization markers (IL-12A, IFN γ 1, CD80, CCL2, CCL4L2, TNF, and IL-23A) were increased, while the anti-inflammatory marker IL-1R2 was downregulated (Figure 4E; Figure S6E). Cholesterol efflux genes (*APOE*, *ABCG1*, and *ABCA5*) were suppressed, and cholesterol uptake genes (*MSR1* [macrophage scavenger receptor 1] and *TLR4*) were upregulated (Figure 4E; Figure S6E). Extracellular matrix remodeling genes were also affected, with downregulation of TIMP1, TIMP3, MMP1, MMP2, MMP3, MMP8, MMP9, MMP14, MMP17, and upregulation of MMP13 (Figure S6B). TF expression was also altered in the miR6855-3p mimic group compared with NC (Figure S6C and S6F).

Together, these transcriptomic analyses demonstrate that elevated miR487a-3p and miR6855-3p drive broad gene expression programs that converge on proinflammatory activation, disrupted lipid metabolism, and extracellular matrix remodeling, constituting hallmarks of proatherogenic macrophage polarization.

miR487a-3p and miR6855-3p Promote Macrophage Proinflammatory Polarization, Foam Cell Formation, and Endothelial Injury

We next determined whether the transcriptomic changes induced by miR487a-3p and miR6855-3p translate

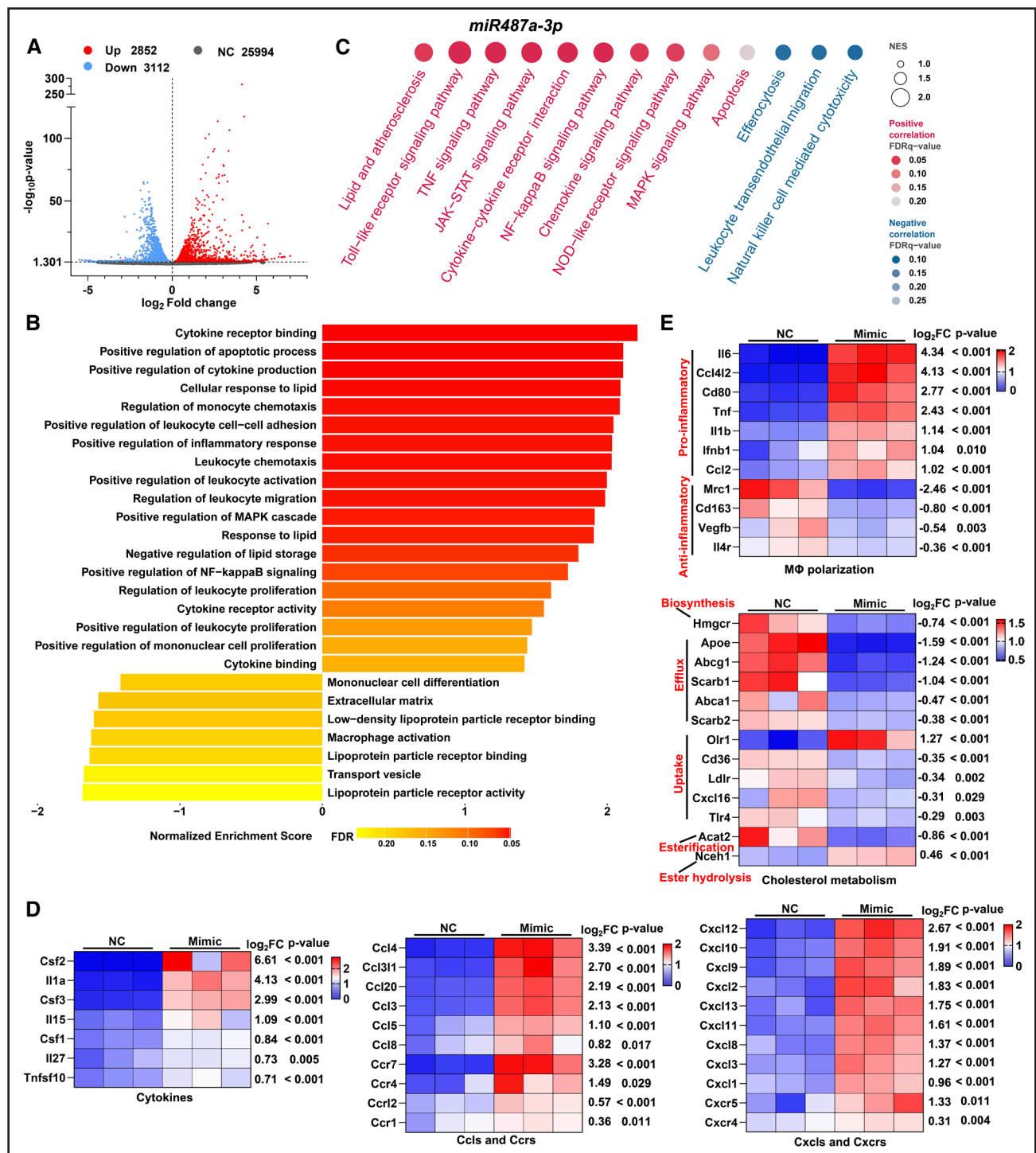


Figure 3. Overexpression of miR487a-3p in THP-1-derived macrophages upregulates signaling related to inflammation and dysregulation of lipid metabolism.

A, Volcano plot showing differentially expressed genes (DEGs) of THP-1 (Tohoku hospital pediatrics-1)-derived macrophages transfected with miR487a-3p mimic or negative control (NC) for 24 hours. **B** and **C**, Gene Set Enrichment Analysis (GSEA) of Gene Ontology (GO) terms (**B**) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways (**C**) between NC and miR487a-3p mimic groups. The top enriched GO terms in the molecular function (MF), biological process (BP), and cellular component (CC) categories are shown in (**B**), and the 13 significantly enriched KEGG pathways are shown in (**C**). **D** and **E**, Heatmaps displaying the expression of DEGs related to cytokines, chemokines, and their receptors (**D**) and macrophage polarization and lipid metabolism (**E**). $n=3$ per group. CCL indicates C-C motif chemokine ligand; CCR, C-C chemokine receptor; CXCL, C-X-C motif chemokine ligand; and CXCR, C-X-C motif chemokine receptor.

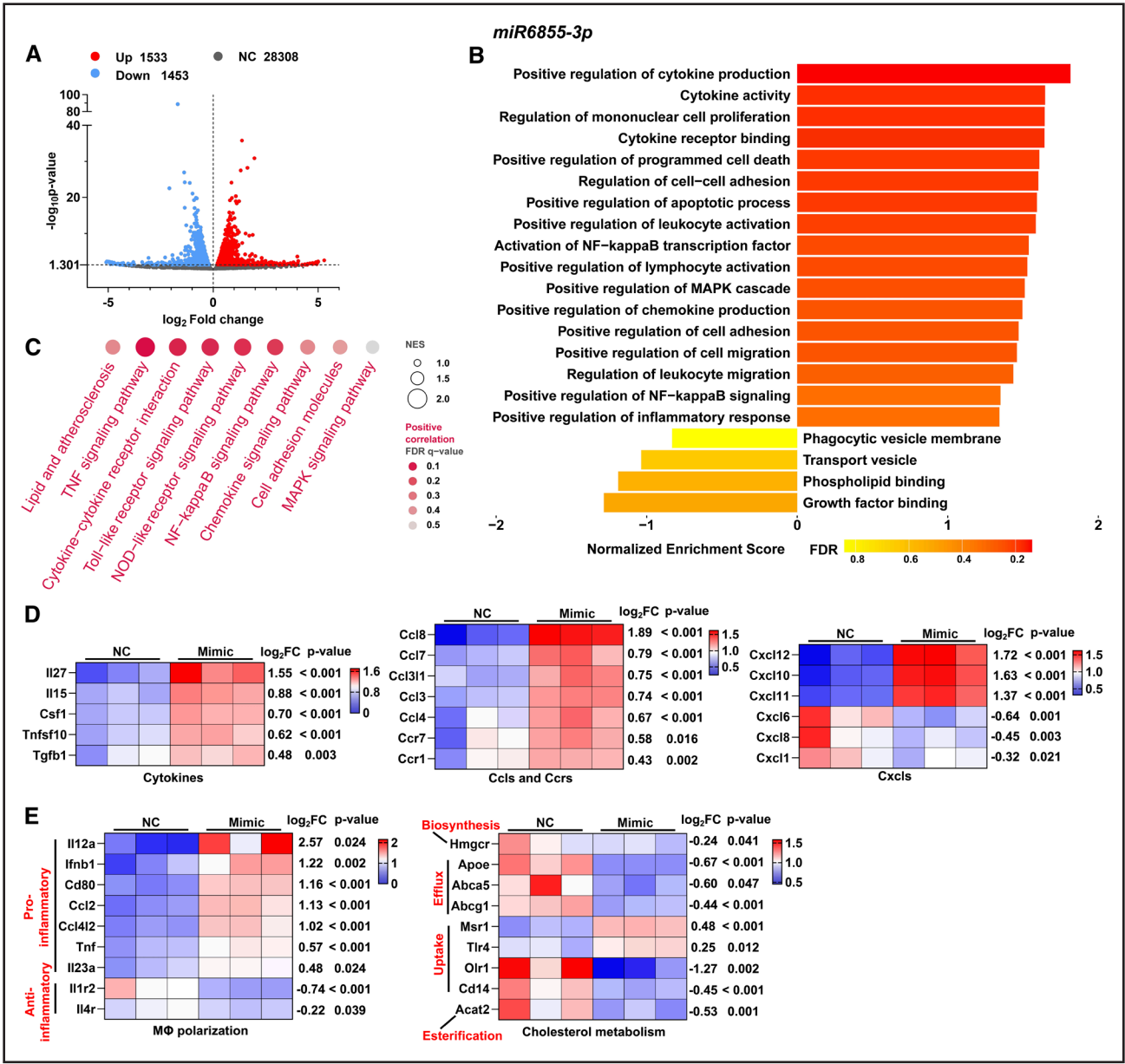


Figure 4. miR6855-3p overexpression drives a shift of gene expression toward inflammation and lipid accumulation in THP-1-derived macrophages.
A, Volcano plot showing differentially expressed genes (DEGs) of THP-1 (Tohoku hospital pediatrics-1)-derived macrophages transfected with miR6855-3p mimic or negative control (NC) for 24 hours. **B** and **C**, Gene Set Enrichment Analysis (GSEA) of Gene Ontology (GO) terms (**B**) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways (**C**) between NC and miR6855-3p mimic groups. The top enriched GO terms in the molecular function (MF), biological process (BP), and cellular component (CC) categories are shown in (**B**), and the 9 significantly enriched KEGG pathways are shown in (**C**). **D** and **E**, Heatmaps displaying the expression of DEGs related to cytokines, chemokines, and their receptors (**D**), and macrophage polarization and lipid metabolism (**E**). n=3 per group. CCL indicates C-C motif chemokine ligand; CCR, C-C chemokine receptor; CXCL, C-X-C motif chemokine ligand; and CXCR, C-X-C motif chemokine receptor.

into functional alterations in macrophage behaviors. miR487a-3p overexpression in THP-1-derived macrophages significantly promoted the secretion of pro-inflammatory cytokines (IL-6, TNF- α , and IL-1 β) and chemokines (CCL2, CCL3, CCL4, CXCL8, CXCL9, and CXCL13) while inhibiting TIMP1 and TIMP2 secretion by >50%, as measured by antibody array analysis (Figure 5A). Flow cytometry confirmed an increase of

the proinflammatory marker CD80 and a concomitant decrease in the anti-inflammatory marker CD206 in miR487a-3p-overexpressing macrophages compared with NC (Figure 5B; Figure S7A). To assess the functional consequence of the polarization on the vascular wall, conditioned medium from miR487a-3p-overexpressing macrophages was applied to human aortic endothelial cells. It significantly induced human aortic endothelial

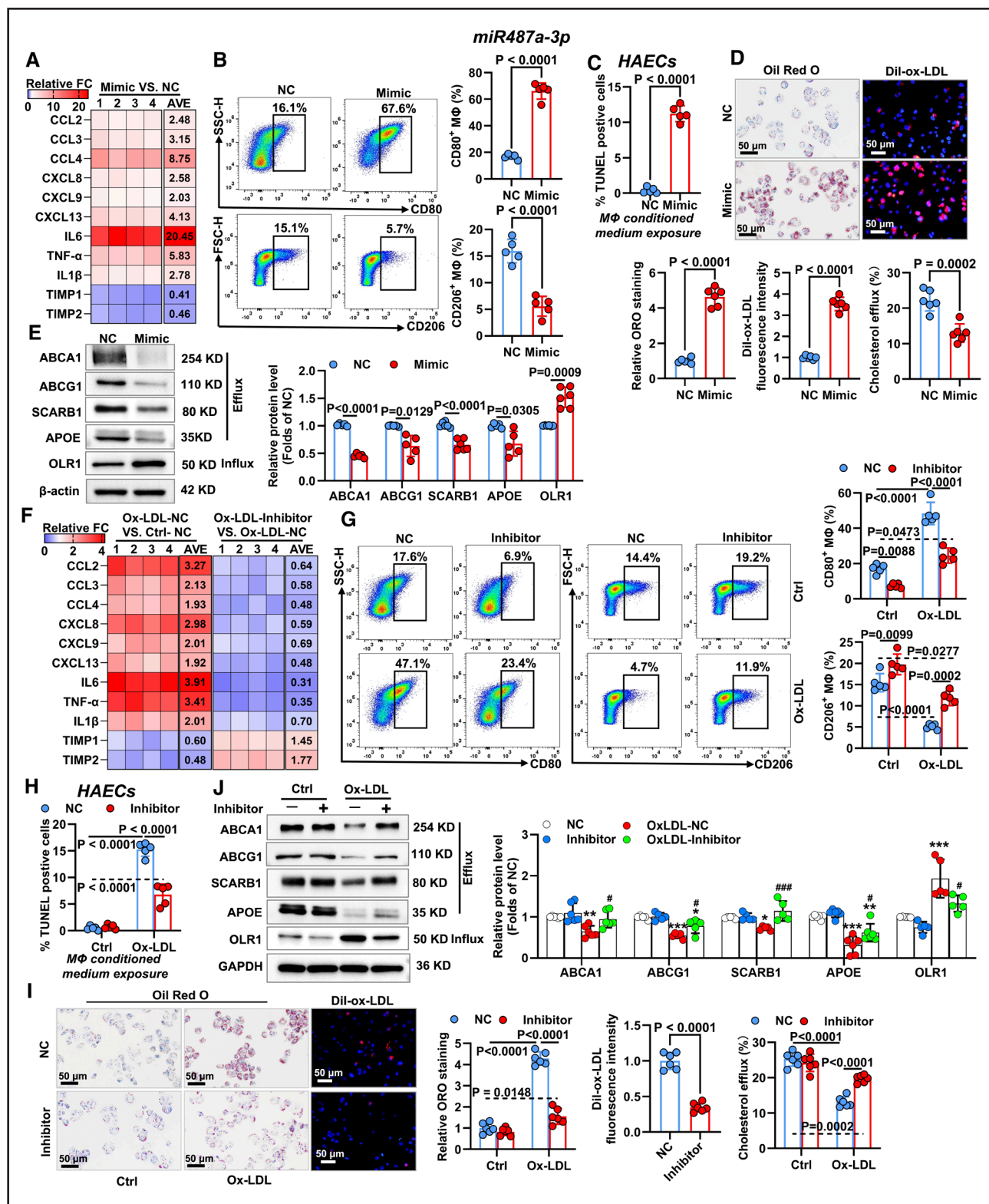


Figure 5. Increases of miR487a-3p mediate macrophages' proinflammatory polarization and foam cell formation in vitro.

A, B, D, and E. THP-1 (Tohoku hospital pediatrics-1)-derived macrophages were transfected with negative control (NC) or miR487a-3p mimic for 24 hours. **F, G, I, and J.** THP-1-derived macrophages were transfected with NC or miR487a-3p inhibitor for 24 hours, followed by incubation with or without 100-μg/mL ox-LDL (oxidized low-density lipoprotein) for an additional 24 hours. **C and H.** Human aortic endothelial cells (HAECs) were treated with macrophage-conditioned medium for 48 hours. **A and F.** Secretion of soluble factors under the indicated treatment was detected by antibody array analysis. The heatmaps indicate the fold change (FC) compared with NC (**A** and **F**) or ox-LDL-NC (**F**). The relative expression levels per replicate and the average FC differences were shown (n=4 biologically independent experiments). **B and G.** Representative flow cytometry scatter plots and quantification of CD80 and CD206 expression in THP-1-derived (Continued)

cell apoptosis compared with treatment with medium from NC macrophages (Figure 5C; [Figure S7B](#)), indicating that miR487a-3p-driven macrophage polarization promotes endothelial injury. Furthermore, miR487a-3p overexpression markedly accelerated foam cell formation by enhancing ox-LDL uptake and inhibiting cholesterol efflux (Figure 5D), accompanied by decreased expression of lipid efflux transporters (ABCA1, ABCG1, SCARB1, and APOE) and increased expression of the ox-LDL receptor OLR1 in macrophages (Figure 5E). These key phenotypes were recapitulated in hMDMs, which similarly exhibited miR487a-3p overexpression-induced proinflammatory polarization, lipid accumulation, and foam cell formation ([Figure S7H through S7K](#)).

Conversely, inhibition of miR487a-3p in THP-1-derived macrophages and hMDMs significantly attenuated ox-LDL-induced miR487a-3p upregulation ([Figure S7C and S7L](#)). miR487a-3p inhibition powerfully suppressed ox-LDL-induced secretion of IL-6, TNF- α , IL-1 β , CCL2, CCL3, CCL4, CXCL8, CXCL9, and CXCL13, restored TIMP1 and TIMP2 secretion, and retarded proinflammatory polarization in both types of macrophages (Figure 5F and 5G; [Figure S7D, S7E, and S7M](#)). Accordingly, endothelial apoptosis induced by conditioned medium from ox-LDL-treated macrophages was significantly blunted by macrophage miR487a-3p inhibition (Figure 5H; [Figure S7F](#)). The ox-LDL-induced foam cell formation was also attenuated by miR487a-3p inhibition through reduced ox-LDL uptake and restored cholesterol efflux capacity (Figure 5I; [Figure S7O](#)), with corresponding recovery of ABCA1, ABCG1, SCARB1, and APOE expression and suppression of OLR1 in 2 types of macrophages (Figure 5J; [Figure S7G and S7N](#)). Together, these data demonstrate that increased miR487a-3p expression contributes to ox-LDL-induced macrophage inflammation, foam cell formation, and secondary endothelial injury.

miR6855-3p overexpression analogously drove THP-1-derived macrophages to secrete TNF- α and chemokines (CCL2, CCL3, and CCL4) while reducing TIMP1 secretion by 20% (Figure 6A). Proinflammatory polarization was confirmed by upregulation of CD80 and downregulation of the anti-inflammatory marker IL-1R2 in both THP-1-derived macrophages and hMDMs (Figure 6B; [Figure S8A, S8H, and S8I](#)). Conditioned medium from miR6855-3p-overexpressing macrophages profoundly induced human aortic endothelial cell apoptosis (Figure 6C; [Figure S8B](#)). miR6855-3p overexpression

also promoted foam cell formation by increasing ox-LDL uptake and impairing cholesterol efflux (Figure 6D; [Figure S8K](#)), with corresponding downregulation of ABCG1 and APOE, and upregulation of the lipid uptake receptors TLR4 and MSR1 in both types of macrophages (Figure 6E; [Figure S8J](#)).

Conversely, miR6855-3p inhibition in THP-1-derived macrophages and hMDMs ([Figure S8C and S8L](#)) significantly reduced ox-LDL-induced secretion of TNF- α , CCL2, CCL3, and CCL4, restored TIMP1 secretion, and attenuated proinflammatory polarization (Figure 6F and 6G; [Figure S8D, S8E, and S8M](#)). Endothelial apoptosis induced by conditioned medium from ox-LDL-treated macrophages was correspondingly reduced by miR6855-3p inhibition (Figure 6H; [Figure S8F](#)). Furthermore, miR6855-3p inhibition suppressed ox-LDL-induced foam cell formation by reducing ox-LDL uptake and restoring cholesterol efflux (Figure 6I; [Figure S8O](#)), with reversal of ox-LDL-induced ABCG1 and APOE downregulation and TLR4 and MSR1 upregulation in both types of macrophages (Figure 6J; [Figure S8G and S8N](#)). Taken together, these results demonstrate that elevated miR6855-3p expression mediates ox-LDL-induced macrophage proinflammatory polarization, lipid accumulation, and consequent endothelial injury.

miR487a-3p and miR6855-3p Directly Target CPE and RRM2, Respectively, to Promote Macrophage Proinflammatory Polarization and Foam Cell Formation

To identify the direct targets mediating the proatherogenic effects of these microRNAs, we performed mRNA sequencing on peripheral monocytes from patients with CAD and HCs (Table 5) and identified 605 DEGs (CAD versus HC; Figure 7A). GO and KEGG analyses revealed enrichment of genes associated with inflammation and lipid metabolism ([Figure S9A and S9B](#)). For each microRNA, we intersected TargetScan-predicted targets with downregulated DEGs from THP-1-derived macrophages (mimic versus NC) and from peripheral monocytes (CAD versus HC). Venn analysis of macrophage DEGs and predicted targets identified 13 candidates for miR487a-3p ([Figure S10A](#)) and 142 candidates for miR6855-3p ([Figure S11A](#)); the top 8 candidates for each were validated by quantitative reverse transcription

Figure 5 Continued. macrophages under the indicated conditions (n=5). **C** and **H**, Quantification of TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labeling) staining in HAECs (n=5). **D** and **I**, Representative images of Oil Red O staining and Dil-ox-LDL uptake assay with the indicated THP-1-derived macrophages and according group quantification, along with the cholesterol efflux assay (n=6). Scale bar, 50 μ m. **E** and **J**, Immunoblotting and quantification of ABCA1 (ATP binding cassette subfamily A member 1), ABCG1 (ATP binding cassette subfamily G member 1), SCARB1 (scavenger receptor class B member 1), APOE (apolipoprotein E), and OLR1 (oxidized low-density lipoprotein receptor 1) in THP-1-derived macrophages with the indicated treatment (n=5–6). GAPDH (glyceraldehyde-3-phosphate dehydrogenase) and β -actin as loading control. Statistics: the unpaired Student *t* test or the Welch *t* test if unequal variance in **B** through **E** and **I**, and the 2-way ANOVA with the post hoc Tukey multiple comparisons between groups in **F** through **J**. **P*<0.05, ***P*<0.01, ****P*<0.001 vs NC, #*P*<0.05, ##*P*<0.01, and ###*P*<0.001 vs ox-LDL-NC in **J**.

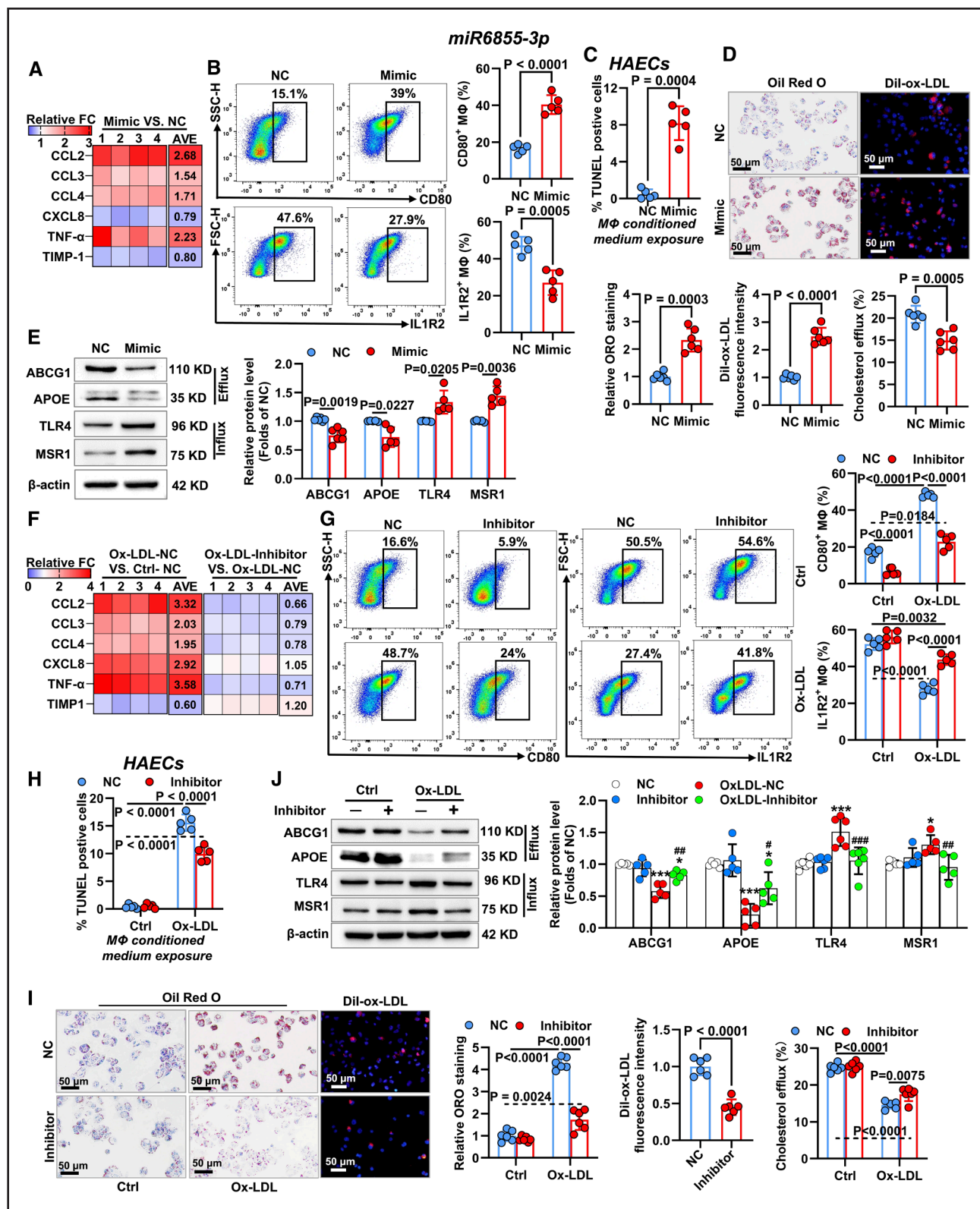


Figure 6. Increases of miR6855-3p facilitate inflammation and lipid accumulation in THP-1-derived macrophages.

A, B, D, and E, THP-1-derived macrophages were transfected with negative control (NC) or miR6855-3p mimic for 24 hours. **F, G, I, and J**, THP-1-derived macrophages were transfected with NC or miR6855-3p inhibitor for 24 hours, followed by incubation with or without 100- μ g/mL ox-LDL (oxidized low-density lipoprotein) for an additional 24 hours. **C and H**, Human aortic endothelial cells (HAECs) were treated with macrophage-conditioned medium for 48 hours. **A and F**, Secretion of soluble factors under the indicated treatment was detected by antibody array analysis. The heatmaps indicate the fold change (FC) compared with NC (**A** and **F**) or ox-LDL-NC (**F**). (Continued)

polymerase chain reaction (Figures S10C and S11C). Intersection of monocyte DEGs with predicted targets yielded 3 candidates for miR487a-3p (Figure S10B) and 42 for miR6855-3p (Figure S11B). A 3-way Venn analysis combining macrophage DEGs, monocyte DEGs, and predicted targets identified CPE as the sole convergent candidate for miR487a-3p (Figure 7B), ranking the highest in both intersections and showing the most significant expression change in miR487a-3p mimic-transfected versus NC macrophages (Figure S10A through S10C). For miR6855-3p, 3 candidates were identified in the same way, among which RRM2 ranked the highest in both intersections and exhibited the most significant expression change (Figure 7B; Figure S11A through S11C).

Dual-luciferase assays in human embryonic kidney-293 cells and THP-1-derived macrophages verified the predicted binding of miR487a-3p to the CPE 3' UTR (untranslated region) and miR6855-3p to the RRM2 3' UTR (Figure 7C). miR487a-3p mimic significantly reduced the luciferase activity of the wild-type CPE 3' UTR without affecting the mutant in both cell types, while miR487a-3p inhibitor restored the luciferase activity of the wild-type CPE 3' UTR that was suppressed by ox-LDL-induced miR487a-3p upregulation in macrophages; neither ox-LDL nor the inhibitor affected the mutant (Figure 7C; Figures S12A and S13A). Similar results were obtained for the binding of miR6855-3p and the RRM2 3' UTR (Figure 7C; Figures S12A and S13E). MicroRNA pulldown assays further confirmed the direct physical interaction: biotin-labeled miR487a-3p markedly enriched CPE mRNA, and biotin-labeled miR6855-3p robustly enriched RRM2 mRNA compared with biotin-miR-NC (Figure 7D). Consistently, CPE and RRM2 mRNA levels were decreased in circulating monocytes of patients with CAD compared with HCs (Figure 7E; Table S4). In THP-1-derived macrophages, miR487a-3p and miR6855-3p mimics significantly reduced CPE and RRM2 mRNA and protein expression, respectively, while ox-LDL treatment alone suppressed both CPE and RRM2, an effect reversed by the corresponding microRNA inhibitors (Figure 7F; Figure S13B and S13F).

CPE is a processing enzyme with established links to metabolic and cardiovascular diseases,^{39–41} and its deficiency has been implicated in inflammatory pathologies. To determine whether CPE mediates the functional

effects of miR487a-3p, we performed rescue experiments in THP-1-derived macrophages. CPE overexpression via lentiviral infection (Figures S12B and S13C) partially reversed miR487a-3p-induced macrophage proinflammatory polarization and foam cell formation (Figure 7G), with decreased expression of inflammatory cytokines and chemokines (CCL3, CCL4, CXCL8, and CXCL13), proinflammatory markers (CD80, CCL2, IL-6, and TNF), and the lipid uptake receptor OLR1, and increased expression of anti-inflammatory markers (CD206 and CD163) and cholesterol efflux transporters (ABCA1 and ABCG1; Figure 7G; Figure S13D). RRM2 is essential for nucleotide metabolism and maintaining dNTP pools,⁴² and its inhibition has been associated with promoting macrophage proinflammatory polarization.^{42–44} Analogously, RRM2 overexpression (Figures S12B and S13G) substantially abrogated miR6855-3p-induced proinflammatory polarization and foam cell formation (Figure 7H), suppressing inflammatory chemokines (CCL3, CCL4, and CXCL8), and proinflammatory markers (CD80, CCL2, and TNF) while enhancing the anti-inflammatory marker IL-1R2 and cholesterol efflux transporters (ABCG1 and APOE; Figure 7H; Figure S13G). Together, these results demonstrate that miR487a-3p and miR6855-3p promote macrophage proatherogenic transformation by directly targeting CPE and RRM2, respectively.

Myeloid-Specific Deletion of CPE or RRM2 Accelerates Atherosclerosis in Hypercholesterolemic Mice

To validate the pathogenic role of the miR487a-3p-CPE and miR6855-3p-RRM2 axes in vivo, we generated mice with monocyte-/macrophage-specific deletion of CPE or RRM2 using the *Lyz2^{CreERT2}* system, thereby mimicking the functional consequence of elevated microRNA levels observed in patients with CAD. Direct microRNA inhibition in mice is precluded by the absence of miR487a-3p and miR6855-3p in the murine genome. This approach is feasible because, among myeloid cells, CPE and RRM2 are expressed primarily in monocytes and macrophages, with negligible expression in neutrophils in *Ldlr* (low-density lipoprotein receptor)^{−/−} mice on either control diet (CD) or high-fat diet (HFD; data not shown). Consistently, CPE⁺ or RRM2⁺ neutrophils constituted <1%

Figure 6 Continued. The relative expression levels per replicate and the average FC differences were shown (n=4 biologically independent experiments). **B** and **G**, Representative flow cytometry scatter plots and quantification of CD80 and IL1R2 expression in THP-1-derived macrophages under the indicated conditions. **C** and **H**, Quantification of TUNEL staining in HAECs. **D** and **I**, Representative images of Oil Red O staining and Dil-ox-LDL uptake assay with the indicated THP-1-derived macrophages and according group quantification, along with the cholesterol efflux assay. Scale bar, 50 μ m. **E** and **J**, Immunoblotting and quantification of ABCG1 (ATP binding cassette subfamily G member 1), APOE (apolipoprotein E), TLR4 (toll-like receptor 4), and MSR1 (macrophage scavenger receptor 1) in THP-1-derived macrophages with the indicated treatment. GAPDH and β -actin as loading control. Data collected from 5 to 6 independent experiments in **B** through **E** and **G** through **I**. Statistics: the unpaired Student *t* test or the Welch *t* test if unequal variance in **B** through **E** and **I**, and the 2-way ANOVA with the post hoc Tukey multiple comparisons between groups in **F** through **J**. **P*<0.05, ****P*<0.001 vs NC, #*P*<0.05, ##*P*<0.01, and ###*P*<0.001 vs ox-LDL-NC in **J**.

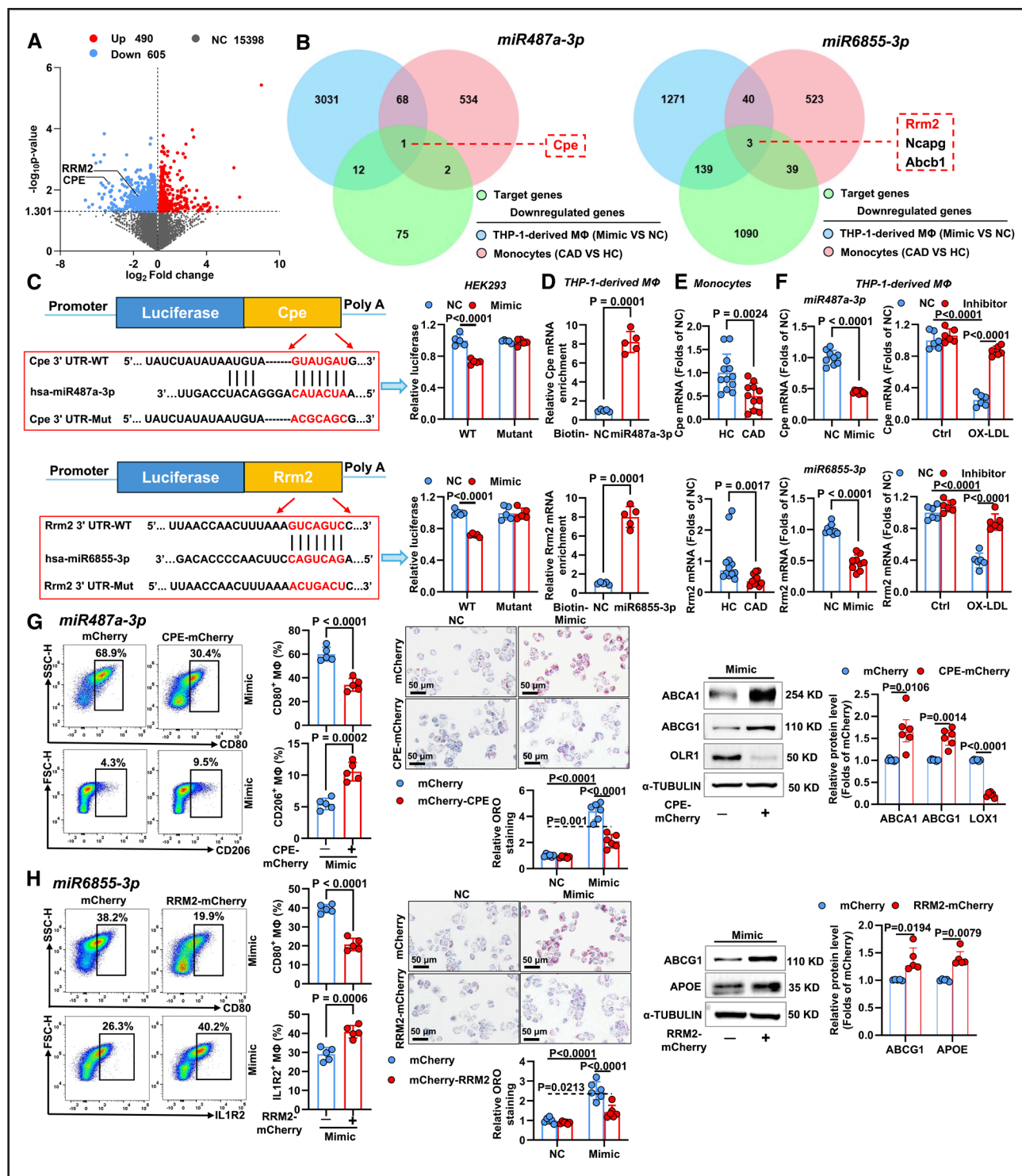


Figure 7. miR487a-3p and miR6855-3p target CPE (carboxypeptidase E) and RRM2 (ribonucleotide reductase regulatory subunit M2), respectively, promoting macrophages' proinflammatory polarization and lipid accumulation.

A, Volcano plot showing differentially expressed genes (DEGs) in peripheral monocytes from healthy controls (HCs; n=10) and patients with coronary artery disease (CAD; n=9). **B**, Venn diagrams depicting the overlap between TargetScan-predicted targets of miR487a-3p or miR6855-3p and downregulated genes in THP-1 (Tohoku hospital pediatric)–derived macrophages (miR487a-3p or miR6855-3p mimic vs negative control [NC]) and in peripheral monocytes (CAD vs HC). **C**, Schematic representation of the predicted binding sites for miR487a-3p and miR6855-3p in the 3' UTR (untranslated region) of CPE and RRM2. Human embryonic kidney (HEK)-293T cells were cotransfected with dual-luciferase reporter vectors containing wild-type (WT) or mutant (MT) CPE or RRM2 3' UTR and with miR487a-3p mimic or 6855-3p mimic or NC. Relative luciferase activity was quantified. **D**, Relative levels of CPE or RRM2 recruited by Biotin-miR487a-3p, Biotin-miR6855-3p, or Biotin-miR-NC probes were measured by miRNA pull-down assay in THP-1-derived macrophages. **E**, The mRNA expressions of CPE and RRM2 in peripheral monocytes from HCs (n=12) and patients with CAD (n=11). (Continued)

of total neutrophils in CPE^{f/f} or RRM2^{f/f} mice (Figure S14A). All the floxed and Lyz2^{CreERT2} mice received adeno-associated virus (AAV) 8-mediated PCSK9 (proprotein convertase subtilisin/kexin type 9) overexpression to induce an Ldlr knockout phenotype and hypercholesterolemia⁴⁵ and were fed CD or HFD containing tamoxifen for 14 weeks (Figure 8A). Tamoxifen treatment extensively depleted CPE⁺ or RRM2⁺ monocytes and macrophages without affecting CPE or RRM2 expression in CPE^{f/f} or RRM2^{f/f} controls (Figure S14A and S14B).

In CPE^{f/f} mice, HFD increased plasma levels of inflammatory cytokines and chemokines (CCL2, CCL3, CCL4, CXCL13, IL-6, and TNF- α), circulating Ly6C^{high} monocytes, and aortic macrophages, all of which were further elevated in CPE^{f/f}Lyz2^{Cre}-HFD mice (Figure 8B and 8C). CPE^{f/f}Lyz2^{Cre}-CD mice also exhibited increased plasma CCL2, IL-6, and TNF- α levels and aortic macrophages compared with CPE^{f/f}-CD mice, whereas peripheral Ly6C^{low} monocyte numbers were unchanged across all groups (Figure S14A). In CPE^{f/f} mice, HFD induced atherosclerotic plaque formation in the whole aorta and aortic root, as well as elevated plasma TC and TG levels, which were further aggravated in CPE^{f/f}Lyz2^{Cre}-HFD mice, except for plasma TG (Figure 8D). Notably, plasma TC levels and aortic root lesion area were also increased in CPE^{f/f}Lyz2^{Cre} mice on CD. As opposed to CPE^{f/f} mice, the HFD-induced suppression of ABCA1 and ABCG1 and upregulation of OLR1 in aortic macrophages were exacerbated by myeloid-specific CPE ablation, which also impaired ABCA1 and ABCG1 expression and increased OLR1 in CD-fed mice (Figure 8E).

Similarly, in RRM2^{f/f} mice, HFD elevated plasma inflammatory cytokines and chemokines (CCL2, CCL3, CCL4, and TNF- α), TC and TG, peripheral Ly6C^{high} monocytes, and aortic macrophages, and promoted atherosclerotic lesion formation, all of which were further intensified in RRM2^{f/f}Lyz2^{Cre}-HFD mice except plasma TG (Figure 8F through 8H). Neither HFD nor myeloid-specific RRM2 depletion affected circulating Ly6C^{low} monocyte numbers (Figure S14A). HFD-induced downregulation of ABCG1 and APOE in aortic macrophages was further exacerbated by RRM2 deficiency, which also reduced ABCG1 and APOE expression and increased plasma TNF- α in CD-fed mice (Figure 8F and 8I).

Together, these in vivo data demonstrate that monocyte-/macrophage-specific deletion of CPE or

RRM2 accelerates atherosclerosis in hypercholesterolemic mice by promoting macrophage inflammatory responses and lipid accumulation, directly recapitulating the in vitro phenotypes induced by miR487a-3p and miR6855-3p overexpression. These findings establish CPE and RRM2 as the key downstream targets of these microRNAs in atherogenesis.

DISCUSSION

We have identified miR487a-3p and miR6855-3p as novel biomarkers and pathogenic mediators in CAD. Both microRNAs were upregulated in peripheral monocytes, peripheral blood mononuclear cells, and coronary plaque macrophages of patients with CAD, with their plasma levels correlating with blood lipids and Gensini score (ROC curve ≈ 0.83 for each), supporting their diagnostic and prognostic value. Their transcription was driven by ox-LDL via KLF5 and IRF1, respectively. Notably, both microRNAs were predominantly expressed in coronary arterial macrophages, and the abundance of miR487a-3p⁺ and miR6855-3p⁺ macrophages was positively associated with plaque area, identifying these macrophage-derived microRNAs as potential therapeutic targets for CAD. In vitro, both microRNAs promoted macrophage proinflammatory polarization, foam cell formation, and endothelial injury in vitro. These effects were mediated through direct targeting of CPE and RRM2, as confirmed by dual-luciferase assays, microRNA pull-down, and functional rescue experiments. Importantly, myeloid-specific deletion of CPE or RRM2 in hypercholesterolemic mice recapitulated the proatherogenic phenotypes, establishing the microRNA-CPE/RRM2 axes as drivers of atherosclerosis progression in vivo.

miR487a-3p has been previously implicated in tumor suppression, autoimmune pathology, and other diseases through multiple reported targets (Table S5). Notably, none of the binding between miR487a-3p and these genes was reported in macrophages (Table S5). Among its 12 known targets, only 4 (AKIP1 [A-kinase interacting protein 1], WNT5A [Wnt family member 5A], CCND1 [cyclin D1], and SOX9 [SRY-box transcription factor 9]) were downregulated in miR487a-3p mimic-transfected versus NC macrophages according to our mRNA sequencing data, and none of these 4 genes showed differential expression in peripheral monocytes

Figure 7 Continued. F, The mRNA expressions of CPE or RRM2 in THP-1-derived macrophages transfected with NC or miR487a-3p mimic or miR6855-3p mimic for 24 hours, or transfected with NC or miR487a-3p inhibitor or miR6855-3p inhibitor for 24 hours followed by treatment with control (Ctrl) or 100- μ g/mL ox-LDL (oxidized low-density lipoprotein) for 24 hours. **G** and **H**, THP-1-derived macrophages were infected with lentiviral vectors containing mCherry-CPE (**G**) or mCherry-RRM2 (**H**) or mCherry and further transfected with NC or miR487a-3p mimic (**G**) or 6855-3p mimic (**H**) for 24 hours. **Left**, Representative flow cytometry scatter plots and quantification of CD80, CD206 (**G**), and IL (interleukin)-1R2 (**H**) expression frequency. **Middle**, Representative images of Oil Red O staining and quantification. **Right**, Immunoblotting and quantitation of ABCG1 (ATP binding cassette subfamily G member 1), ABCA1 (ATP binding cassette subfamily A member 1; **G**), OLR1 (oxidized low-density lipoprotein receptor 1; **G**), APOE (apolipoprotein E; **H**), and α -tubulin as loading control. Data collected from 5 to 6 independent experiments in **C**, **D**, and **F** through **H**. Statistics: the 2-way ANOVA with the post hoc Tukey multiple comparisons between groups in **C** and **F**. The unpaired Student *t* test in **E** through **H**, the Welch *t* test in **D** and **F** through **H**, and the Mann-Whitney *U* test in **E** and **H**.

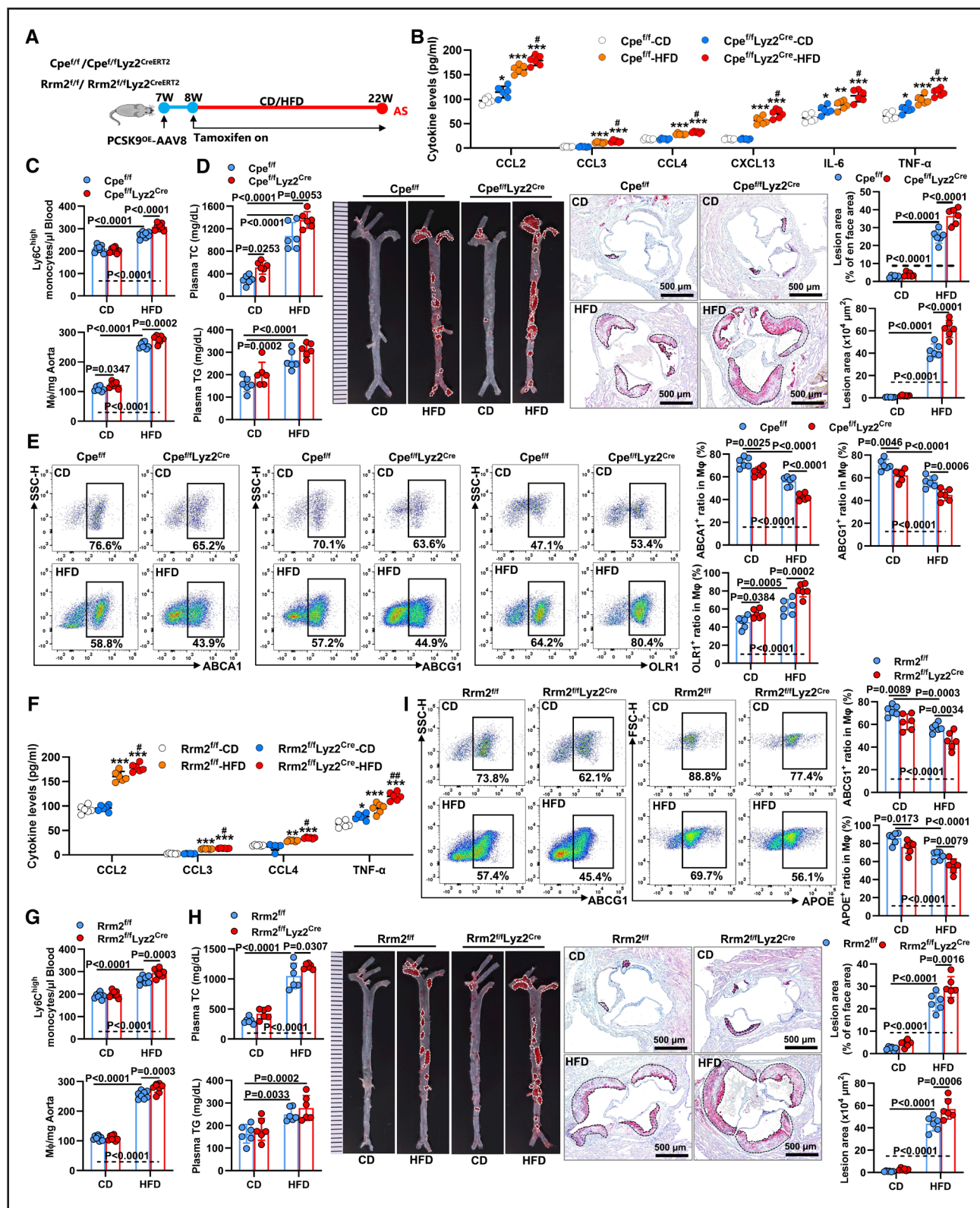


Figure 8. Myeloid-specific CPE (carboxypeptidase E) or RRM2 (ribonucleotide reductase regulatory subunit M2) depletion aggravates inflammation and promotes atherosclerosis development in PCSK9 (proprotein convertase subtilisin/kexin type 9) overexpression mice on a high-fat diet (HFD).

A, Protocol for myeloid-specific CPE or RRM2 ablation in $Cpe^{fl/fl}$ or $Rrm2^{fl/fl}$ -Lyz2^{CreERT2} mice ($Cpe^{fl/fl}$ or $Rrm2^{fl/fl}$ mice as controls) with tamoxifen administration in control diet (CD) or HFD from 8 to 22 weeks after 1 week of PCSK9^{OE}-AAV8 (adeno-associated virus 8) infection. **B** and **F**, Plasma levels of CCL (C-C motif chemokine ligand) 2, CCL3, CCL4, CXCL (C-X-C motif chemokine ligand) 13, IL (interleukin)-6, and TNF (tumor necrosis factor)- α in myeloid-specific CPE (**B**) or RRM2 (**F**) deletion mice ($n=6$ per group). (*Continued*)

between patients with CAD and controls (Figure S10D), indicating that they are unlikely to mediate miR487a-3p function in the context of CAD. In addition to these known functions, our study reveals a proinflammatory and proatherogenic role for miR487a-3p in macrophages, where it targets CPE to promote inflammation and lipid dysregulation, thereby accelerating atherosclerosis. Similarly, miR6855-3p was previously reported to promote antioxidant protection in breast cancer cells via noncanonical translational activation of PRDX5A (peroxiredoxin 5A).⁴⁶ Although PRDX5A is expressed in macrophages and implicated in redox-sensitive inflammatory signaling,⁴⁷ PRDX5A expression did not differ between miR6855-3p mimic and NC groups in macrophages, nor between CAD and HC groups in peripheral monocytes (Figure S11D). Instead, elevated miR6855-3p promoted macrophage proinflammatory polarization and foam cell formation through canonical 3' UTR targeting of RRM2. These findings underscore the context-dependent pleiotropy of microRNAs, wherein a single microRNA can drive divergent, even opposing, outcomes depending on cell type, target gene repertoire, and disease milieu.

The elevated plasma levels of miR487a-3p and miR6855-3p likely reflect their release from activated monocytes and macrophages, as demonstrated by our in vitro secretion assays. Circulating microRNAs are remarkably stable due to their association with extracellular vesicles, Argonaute proteins, and lipoproteins^{1,28,48} and can mediate intercellular communication.^{49,50} The positive correlation of both microRNAs with the Gensini score and their robust ROC performance support their utility as minimally invasive biomarkers for CAD diagnosis and prognosis. Importantly, multivariate logistic regression analysis demonstrated that both microRNAs remained significant independent predictors of CAD after adjustment for TC, LDL-C, and sdLDL-C, whereas the lipid parameters themselves lost significance. This may reflect the higher prevalence of cholesterol-lowering medication use in the CAD group (Table 2), which reduces circulating lipid levels without necessarily reversing the microRNA upregulation initially driven by hypercholesterolemia. These findings support the potential utility of these microRNAs as stable biomarkers of atherogenic burden that complement traditional lipid profiling.

Monocytes infiltrate the arterial intima in response to endothelial injury, differentiate into macrophages, and

drive plaque progression through local proliferation.^{51,52} We found that miR487a-3p⁺ and miR6855-3p⁺ macrophages constituted ≈60% of total plaque macrophages in patients with CAD, and their abundance positively correlated with lesion area. Based on our functional data, we postulate that aberrant upregulation of these microRNAs in circulating monocytes promotes their proinflammatory activation and recruitment into plaques, while sustained microRNA expression in lesional macrophages drives foam cell formation and inflammatory cytokine secretion, amplifying atherogenesis. Inhibition of either microRNA substantially attenuated ox-LDL-induced inflammatory responses and lipid accumulation, supporting their potential as therapeutic targets. In addition, both microRNAs are lipid-inducible, with their transcription mediated by KLF5 and IRF1, respectively, providing a mechanistic link between hyperlipidemia and microRNA-driven macrophage activation in CAD.

CPE, a prohormone processing enzyme linked to metabolic and cardiovascular diseases,^{39–41} was suppressed by miR487a-3p in monocytes and macrophages. CPE has been shown to exert anti-inflammatory effects in the intestine by modulating cytokine levels and myeloid cell migration,⁵³ and its overexpression protects cardiomyocytes from ischemia/hypoxia injury.³⁹ Consistent with these observations, CPE overexpression mitigated miR487a-3p-induced inflammation and restored lipid metabolic homeostasis in our study. Importantly, myeloid-specific CPE deficiency exacerbated inflammation, impaired cholesterol efflux transporter expression, and accelerated atherosclerosis in hypercholesterolemic mice, indicating a protective role for CPE in atherogenesis. RRM2, essential for dNTP synthesis and genome maintenance,⁴² has been reported to regulate macrophage polarization, with its inhibition promoting proinflammatory phenotypes.^{43,44} We found that miR6855-3p suppressed RRM2 expression, and RRM2 overexpression reversed miR6855-3p-induced proinflammatory polarization and foam cell formation. Conversely, myeloid-specific RRM2 deficiency exacerbated inflammation and atherosclerosis in vivo. These results establish RRM2 as a critical regulator of macrophage lipid metabolism and inflammatory homeostasis. Together, these findings identify CPE and RRM2 as functionally convergent downstream effectors of miR487a3p and miR6855-3p, respectively, whose suppression coordinates macrophage inflammatory activation and lipid metabolic dysfunction in atherosclerosis.

Figure 8 Continued. **C** and **G**, Quantitation of blood Ly6C^{high} monocytes and aortic macrophages in myeloid-specific CPE (**C**) or RRM2 (**G**) knockout mice (n=8 per group). **D** and **H**, Plasma lipid profiles and atherosclerotic lesion analysis in myeloid-specific CPE-deficient (**D**) or RRM2-deficient (**H**) mice (n=6 per group). **Left**, Plasma total cholesterol (TC) and triglyceride (TG) levels. **Middle**, Representative Oil Red O (ORO) staining of en face whole aortas and aortic root cross-sections; lesion areas were outlined by dashed lines. **Right**, Quantitative evaluation of the percentage of ORO-positive plaque area in whole aortas and absolute lesion area in the aortic roots. **E** and **I**, Flow cytometry plots and quantitation of ABCG1 (ATP binding cassette subfamily G member 1), ABCA1 (ATP binding cassette subfamily A member 1; **E**), OLR1 (oxidized low-density lipoprotein receptor 1; **E**), and APOE (apolipoprotein E; **I**) expressions in aortic macrophages from mice harboring myeloid-specific deletion of CPE (**E**) or RRM2 (**I**; n=6 per group). Statistics: the 2-way ANOVA with the post hoc Tukey multiple comparisons between groups in **B** through **I**. **P*<0.05, ***P*<0.01, and ****P*<0.001 vs CPE^{fl/fl}-CD (**B**) or RRM2^{fl/fl}-CD (**F**), and #*P*<0.05, ##*P*<0.01, and ###*P*<0.001 vs CPE^{fl/fl}Lyz2^{Cre}-CD (**B**) or RRM2^{fl/fl}Lyz2^{Cre}-CD (**F**).

Our results must be considered in the context of potential study limitations. First, validation in larger, independent cohorts is needed to confirm the clinical utility of these microRNAs as biomarkers and to develop robust multivariable prediction models. Second, microRNA mimic transfection achieved supraphysiological levels although key findings were corroborated by inhibitor rescue and in vivo target gene deletion. Last, direct in vivo microRNA inhibition would further strengthen the translational relevance; this represents an important future direction that would require humanized microRNA mouse models. In addition, our in vivo studies were conducted exclusively in male mice to avoid confounding by the female estrous cycle; future studies in female mice are warranted to assess potential sex-specific differences in the miR487a-3p-CPE and miR6855-3p-RRM2 axes in atherosclerosis.

In summary, miR487a-3p and miR6855-3p, through targeting CPE and RRM2, respectively, drive macrophage proinflammatory responses and lipid metabolic dysregulation in atherosclerosis. Their release into the circulation, combined with their correlation with CAD severity, establishes these microRNAs as novel biomarkers for CAD diagnosis and prognosis, while their pathogenic mechanisms identify them as potential therapeutic targets.

ARTICLE INFORMATION

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Disclosures

None.

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